arXiv:2604.23996v2 [cs.CV] 28 Sep 2026

Soft Modality-Guided Expert Specialization in MoE-VLMs

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Abstract

Mixture-of-Experts (MoE) has become a prevalent backbone for large vision-language models (VLMs), yet how modality-specific signals should guide expert routing remains under-explored. Existing routing strategies are either hand-crafted or modality-agnostic, relying on idealized priors that ignore the layer-dependent modality fusion patterns in MoE-VLMs and provide little guidance for expert specialization. We propose Soft Modality-guided Expert Specialization (SMoES), which consists of dynamic soft modality scores that capture layer-dependent fusion patterns, an expert binning mechanism aligned with expert-parallel deployment, and an inter-bin mutual information regularization that encourages coherent modality specialization. Our method leverages attention-based or Gaussian-statistics modality scores to optimize mutual information regularization. Experiments across four MoE-based VLMs and 16 benchmarks demonstrate improvement on both effectiveness and efficiency: 0.9% and 4.2% average gain on multimodal and language-only tasks, 56.1% reduction in EP communication overhead, and 12.3% throughput improvement under realistic deployment. These results validate that aligning routing with modality-aware expert specialization unlocks MoE-VLM capacity and efficiency.

1. Introduction

The Mixture-of-Experts (MoE) architecture [14, 19, 27, 45] has emerged as a cornerstone for modern large vision-language models (VLMs). Its principle of conditional computation allows for a massive expansion of model capacity with only a modest increase in the per-token computational budget by routing inputs to specialized expert subnetworks [26, 45]. This paradigm is particularly well-suited for fusing heterogeneous modalities. Consequently, leading systems like DeepSeek-VL2 [60], Kimi-VL [51], GLM-4.5V [22], and InternVL-3.5 [56] have adopted MoE to achieve better performance while maintaining tractable inference costs. Despite this widespread adoption, a fundamental question remains under-explored: how do modality-specific signals (vision vs. text) interact with the expert routing mechanism, and can it be optimized to enhance both effectiveness and efficiency?

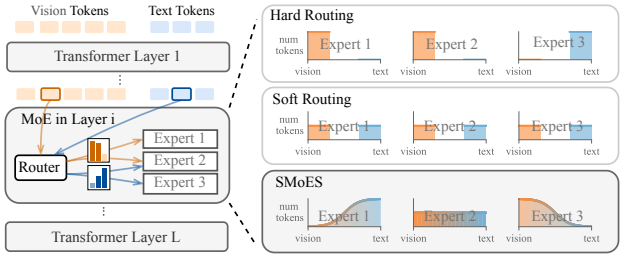

Figure 1. Comparison of routing strategies in MoE-VLMs. Hard routing enforces strict modality-expert separation, while soft routing allows arbitrary, uncontrolled mixing. SMoES uses soft modality scores to encourage dynamic expert specialization.

Mainstream MoE-VLMs route tokens using either hard or soft routing paradigms (Fig. 1). Hard routing [2, 55] pre-assigns experts to a specific modality, creating sharp specialization at the cost of rigid boundaries that hinder adaptation to cross-modal features and ignore the natural blending of representations across layers. In contrast, soft routing [22, 42, 51, 60], the prevailing paradigm, allows experts to process any token. Yet, these methods often rely on heuristic priors or auxiliary losses disconnected from the evolving modality distributions, resulting in either over-mixing or under-specialization. Hybrid approaches [2, 53, 55] partition experts into modality-specific (hard-routed) and shared (soft-routed) groups, but this partitioning is typically hand-crafted and layer-agnostic, failing to align with feature evolution. Fundamentally, these paradigms are guided by idealized assumptions or uncontrolled mixing rather than the dynamic, data-driven geometry of modality distributions, offering little guidance on how expert specialization should evolve with network depth.

We analyze modality fusion patterns of token features across layers in LLaVA-1.5 [35] and a DeepSeekMoE-based [11] VLM, as shown in Fig. 2. The analysis shows that modality fusion in MoE-VLMs is highly heterogeneous, with no clear boundary between modality-specific and cross-modal states. At a macro level, fusion pat-

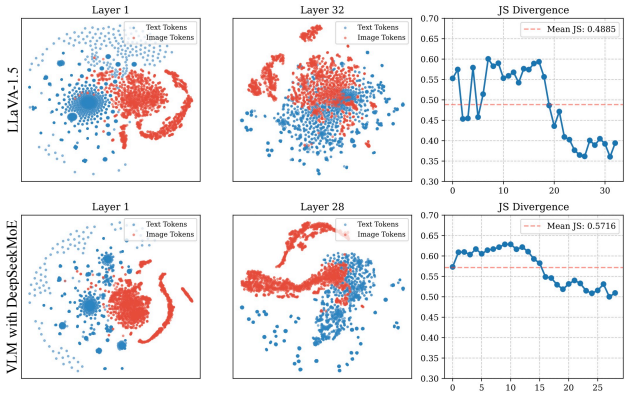

Figure 2. Modality fusion pattern. Token feature distributions vary across models and layers. Text tokens are divided into several semantic clusters, while vision tokens are roughly separated into content and padding clusters. The last column shows the JS divergence between vision and text tokens across all layers.

terns vary across models and layers, as reflected by distinct Jensen–Shannon (JS) divergence trajectories. At a micro level, even within the same layer and modality, some tokens remain modality-specific while others become cross-modal. This multi-scale heterogeneity indicates that rigid routing assumptions—either enforcing hard separation or imposing uniform mixing—are misaligned with how modalities actually interact across depth. Therefore, guided by signals that respect the evolving modality structure, the expert pool in MoE has the potential to naturally enable dynamic modality specialization.

Beyond effectiveness, modality specialization is also closely tied to the inference efficiency of MoE-VLMs. Vision and text tokens differ sharply in both quantity and information density. Vision tokens often dominate the sequence length but typically carry lower information density due to spatial redundancy, whereas text tokens are fewer yet more semantically concentrated [5, 24, 40]. In real-world scenarios, inputs may even be text-only. This asymmetry poses two efficiency challenges. First, standard routing objectives coupled with vanilla load balancing tend to allocate most experts to the dominant yet low-information-density vision modality, hindering optimal specialization. Second, under expert parallelism (EP) [34]—a common deployment strategy—modality-agnostic routing scatters tokens across devices, inflating inter-communication overhead. Carefully-designed modality specialization addresses both issues: by establishing clear expert–modality affinities and scheduling tokens accordingly, aligned experts can be co-located on the same device, reducing communication while preserving balanced computation and capacity.

Motivated by these observations, we propose Soft Modality-guided Expert Specialization (SMoES), which addresses these challenges through three key components. First, we introduce dynamic soft modality scores through two complementary estimators—attention-accumulated and Gaussian-statistics—that enable adaptive expert selection aligned with actual fusion states rather than hard boundaries. Second, we introduce expert binning—partitioning experts into coherent groups—which serves as the structural foundation for modality specialization and enables natural device placement units for modality-aware expert parallelism deployment. Third, to drive effective modality specialization, we introduce a mutual information (MI) objective that encourages different expert bins to specialize on distinct modality patterns.

In summary, we make three key contributions: (1) Dynamic soft modality scores—Gaussian-statistics and attention-accumulated estimators—that align routing with the evolving modality fusion patterns across layers. (2) An expert binning mechanism coupled with an MI-driven specialization objective that promotes coherent modality grouping for both effectiveness and deployment efficiency. (3) Extensive experiments across four MoE-VLM backbones and 16 benchmarks demonstrating SOTA performance in accuracy and efficiency, including reduced EP communication overhead in realistic deployment.

2. Related Works

2.1. Mixture-of-Experts

Sparsely-gated MoE architectures enable conditional computation through sparse activation and efficient gating [14, 27, 45]. Recent works like DeepSeek-V3 further improve routing efficiency, load balancing, and communication performance [34, 43, 68]. Mutual-information (MI) constraints have been explored to regulate expert selection. For expert-task alignment, Mod-Squad and MTHL maximize MI between tasks and experts [7, 8]. For expert-token alignment, ModuleFormer and CoMoE maximize MI between tokens and modules through entropy balancing or contrastive learning [15, 48]. However, these methods overlook modality differentiation in multimodal models. Recent VLMs increasingly adopt MoE backbones [18, 22, 51, 52, 56, 60]. Related approaches also include LoRA-based expert expansion [6, 46, 59] and converting dense VLMs to sparse MoE architectures [25, 31, 67].

2.2. Modality Specialization in MoE-based VLMs

For modality-aware routing, existing methods can be categorized by their degree of expert-modality coupling. **Hard routing** assigns experts exclusively to single modalities [1, 13, 29, 30, 33, 39, 47], achieving strong specialization but sacrificing flexibility across modalities. **Hybrid routing** allows certain experts to handle multiple modalities while others remain modality-specific [2, 53, 55], yet relies on manual partitioning misaligned with actual fusion dynamics. **Naive soft routing** [4, 42] permits flexible expert mixing without explicit modality guidance, leaving specializa-

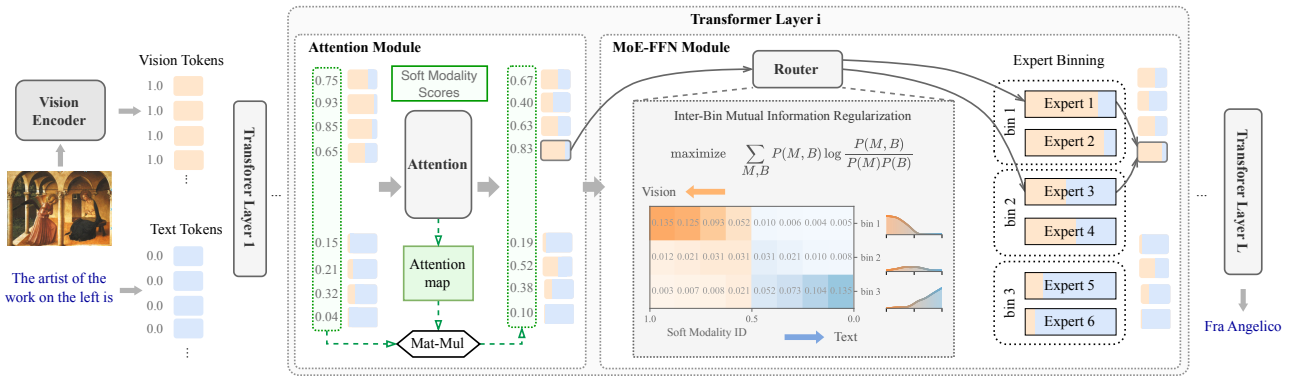

Figure 3. Overview of SMoES. **Soft Modality Scores**: Tokens start with hard modality IDs (0/1 for vision/text) and are progressively refined into layer-wise soft scores $M \in [0, 1]$. We estimate them via (i) attention-accumulated (depicted here) or (ii) Gaussian-statistics estimation (detailed in Fig. 4). **Expert Binning**: experts learn bin-level modality preference for deployment efficiency. **Mutual Information** Regularization: maximizes mutual information between soft modality scores (M) and bins (B), inducing modality preference.

tion to implicit optimization. **Smarter soft routing** introduces auxiliary objectives to shape routing: LTDR balances expert load on long-tailed vision distributions [3], and STGC mitigates gradient conflicts within experts [63]. SMAR achieves modality differentiation by using KL divergence to regularize routing distributions toward modality-specific patterns [61]. In contrast to KL-based regularization, our approach employs mutual information constraints between modalities and experts to guide specialization. Moreover, we introduce layer-adaptive soft modality scores that capture dynamic fusion patterns across layers, and expert binning for efficient distributed deployment.

2.3. Efficient Deployment of MoE Models

As MoE models scale, deployment efficiency becomes increasingly critical, with load imbalance and All-to-All communication overhead as key challenges in distributed expert-parallel (EP) settings [66]. To mitigate load imbalance, MoGE enforces balanced expert activation through predefined groups [50], while Grove-MoE explores heterogeneous expert sizes that dynamically adapt to token complexity [58]. Complementary strategies address the straggler effect: Expand Drop employs capacity-aware token dropping to limit expert overload [20], and AEP decouples layer execution with asynchronous queuing and adaptive re-batching [54]. However, for MoE-VLMs, how to leverage modality fusion characteristics to guide expert partitioning and communication strategies under EP deployment remains largely underexplored.

3. Method

Modality fusion in MoE-VLMs exhibits layer-varying patterns with smooth token-level transitions, while vision/text asymmetry creates load imbalance and inflates EP communication overhead. To address these challenges, we introduce SMoES, which integrates three components: soft modality scores capturing continuous token-level modality patterns, expert binning creating modality-aligned partitions, and inter-bin mutual information regularizing coherent specialization, as illustrated in Fig. 3.

3.1. Preliminaries: MoE in VLM

A typical MoE-based VLM consists of a vision encoder, projection layer, and an MoE-augmented LLM backbone. Within each MoE layer l, token features $\mathbf{x}_{ij} \in \mathbb{R}^D$ (where i indexes the sample and j indexes the token) are routed by a gating network to a pool of N_e experts. The router computes gating scores $g_{ij,e} = \text{softmax}(\mathbf{W}_{\text{gate}}\mathbf{x}_{ij})_e$, and standard top-$k$ routing selects the k highest-scoring experts. To prevent routing collapse, a load-balancing auxiliary loss [14, 45] is typically applied:

$$\mathcal{L}_{\text{bal}} = \sum_{l} N_e \sum_{e=1}^{N_e} f_e \, P_e \tag{1}$$

where f_e is the fraction of tokens routed to expert e, and P_e is the average gating score.

3.2. Soft Modality Scores

The multi-scale heterogeneity of modality fusion (Fig. 2) reveals smooth, layer-dependent transitions in token representations rather than fixed modality identities. Hard modality indicators—binary labels assigned at the input—fail to capture these continuous fusion dynamics. To guide routing aligned with evolving modality patterns, we introduce *soft modality scores* $M^{(l)}_{ij,m} \in [0, 1]$ for each token and modality $m \in \{\text{text}, \text{vision}\}$ at layer l, where $\sum_m M^{(l)}_{ij,m} = 1$. We develop two complementary estimators: an *attention-accumulated score* that captures local cross-token interactions; and a *Gaussian-statistics score* that captures global statistical regularities across the dataset.

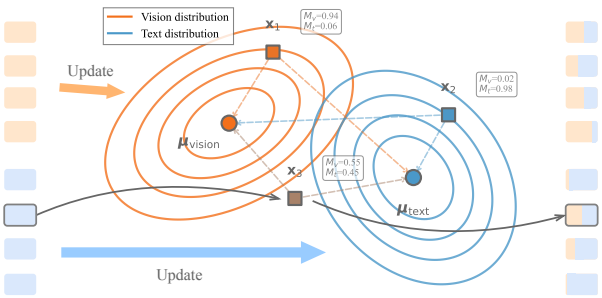

Figure 4. **Gaussian-statistics score estimation.** Per-modality Gaussian distributions are continuously updated from token batches. For each token, we compute modality affinity as soft scores based on log-likelihood under each distribution.

3.2.1. Attention-Accumulated Score

The attention mechanism provides a natural pathway for modality signal propagation: when a token attends to others, it absorbs their modality characteristics proportionally to attention weights. We initialize scores at layer 0 with hard modality indicators:

$$M_{ij,m}^{\text{attn},(0)} = \mathbf{1}\{m = m(\mathbf{x}_{ij})\}. \tag{2}$$

For subsequent layers, we update in two steps. First, aggregate attended tokens' scores using attention weights:

$$\tilde{M}_{ij,m}^{\text{attn},(l)} = \sum_{j'=1}^{J} \text{Attn}_{j,j'}^{(l)} \cdot M_{ij',m}^{\text{attn},(l)} \tag{3}$$

where $\text{Attn}^{(l)}$ is the layer l attention matrix averaged across heads, and J is the sequence length. Second, combine aggregated and input scores via residual-weighted update:

$$M_{ij,m}^{\text{attn},(l+1)} = \frac{\|\mathbf{x}_{\text{attn},ij}^{(l)}\| \cdot \tilde{M}_{ij,m}^{\text{attn},(l)} + \|\mathbf{x}_{ij}^{(l)}\| \cdot M_{ij,m}^{\text{attn},(l)}}{\|\mathbf{x}_{\text{attn},ij}^{(l)}\| + \|\mathbf{x}_{ij}^{(l)}\|} \tag{4}$$

This mirrors the residual structure of Transformers ($\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)} + \mathbf{x}_{\text{attn}}^{(l)}$), where feature norms weight the contributions of attention and residual paths.

3.2.2. Gaussian-Statistics Score

Complementary to the attention-accumulated score's local, sequence-specific perspective, we propose the *Gaussian-statistics score*—a global, distribution-based estimator. The key insight is that tokens from different modalities exhibit distinct feature distributions in the embedding space. We exploit these distributional differences to instantaneously infer modality affiliation at each layer, independent of Layer 0 initialization.

For each layer l and modality m, we maintain a Gaussian distribution with diagonal covariance: mean $\boldsymbol{\mu}_m \in \mathbb{R}^D$ and variance $\boldsymbol{\sigma}_m^2 \in \mathbb{R}^D$. Statistics are updated online via exponential moving average (EMA) variant of Welford's algorithm [57]:

$$N_{m,t} = \beta N_{m,t-1} + |\mathcal{T}_m| \tag{5}$$

$$\mathbf{S}_{\mu,m,t} = \beta \mathbf{S}_{\mu,m,t-1} + |\mathcal{T}_m| \cdot \boldsymbol{\mu}_{\mathcal{T}_m} \tag{6}$$

$$\mathbf{S}_{\sigma^2,m,t} = \beta \mathbf{S}_{\sigma^2,m,t-1} + \sum_{\mathbf{x}\in\mathcal{T}_m} (\mathbf{x} - \boldsymbol{\mu}_{\mathcal{T}_m})^2 + (\boldsymbol{\mu}_{\mathcal{T}_m} - \tfrac{\mathbf{S}_{\mu,m,t-1}}{N_{m,t-1}})^2 \cdot \tfrac{\beta N_{m,t-1}|\mathcal{T}_m|}{\beta N_{m,t-1}+|\mathcal{T}_m|} \tag{7}$$

where $\mathcal{T}_m$ is the set of modality-m tokens in the current batch, $\boldsymbol{\mu}_{\mathcal{T}_m}$ is the batch mean, and β is the EMA decay factor. The distribution parameters at time t are: $\boldsymbol{\mu}_m = \mathbf{S}_{\mu,m,t}/N_{m,t}$ and $\boldsymbol{\sigma}_m^2 = \mathbf{S}_{\sigma^2,m,t}/N_{m,t}$.

To infer each token's modality affiliation, we compute its log-likelihood under each distribution:

$$\text{LL}_{ij,m} = -\frac{1}{2}\sum_{d=1}^{D}\left(\log \sigma_{m,d}^2 + \frac{(x_{ij,d} - \mu_{m,d})^2}{\sigma_{m,d}^2}\right) \tag{8}$$

The soft modality score is obtained via temperature-scaled softmax:

$$M_{ij,m}^{\text{gauss}} = \frac{\exp(\text{LL}_{ij,m}/\tau)}{\sum_{m'\in\{\text{text,vision}\}} \exp(\text{LL}_{ij,m'}/\tau)} \tag{9}$$

where temperature τ controls score sharpness.

3.3. Expert Binning

In expert-parallel (EP) deployments, modality-agnostic routing scatters tokens across devices, amplifying communication overhead. To reduce this, we co-locate experts with similar modality preferences on the same device. Moreover, real deployments vary in device count, requiring a flexible binning mechanism.

We introduce *expert binning*: at each layer, partition experts into N_{bins} groups $\mathbf{B} = \{\mathbf{B}_1, \ldots, \mathbf{B}_{N_{\text{bins}}}\}$ with $N_B = N_e/N_{\text{bins}}$ experts each, where N_{bins} can match device count. By encouraging bins to develop distinct modality preferences through inter-bin MI (detailed below), we enable modality-aligned placement: bins with similar preferences co-locate, reducing communication while balancing load.

To form bins reflecting modality preferences, we adopt *momentum-adaptive binning*. We track each expert's load from each modality using EMA:

$$\bar{C}_{m,e,t} = \beta \bar{C}_{m,e,t-1} + (1-\beta) C_{m,e} \tag{10}$$

where $C_{m,e} = |\{(i,j) \in \mathcal{T}_m \mid e \in \text{TopK}(g_{ij})\}|$ is the count of modality-m tokens routed to expert e. We compute each expert's text-bias score:

$$f_{\text{spec}}(e) = \frac{\bar{C}_{\text{text},e}}{\bar{C}_{\text{text},e} + \bar{C}_{\text{vision},e}} \tag{11}$$

Experts are sorted by f_{spec} and partitioned into N_{bins} consecutive bins, grouping experts with similar modality preferences together.

3.4. Inter-Bin Mutual Information Regularization

We introduce an *inter-bin MI objective* to drive modality differentiation among expert bins. The intuition is that high MI between modality M and bin $\mathbf{B}$ means knowing the selected bin provides strong information about token modality, thereby encouraging specialized bins. Mathematically, we maximize $I(M;\mathbf{B})$.

To compute MI, we first compute the average gating score for each sample i, modality m, and bin $\mathbf{B}_k$:

$$\bar{S}_{i,m,\mathbf{B}_k} = \frac{\sum_{e\in\mathbf{B}_k}\sum_j M_{ij,m}\cdot g_{ij,e}}{N_B\sum_j M_{ij,m}} \tag{12}$$

where $M_{ij,m}$ is the soft modality score and $g_{ij,e}$ is the gating score. We treat this as an unnormalized distribution and compute the normalized joint probability:

$$P_i(m,\mathbf{B}_k) = \frac{\bar{S}_{i,m,\mathbf{B}_k}}{\sum_{m'}\sum_{k'}\bar{S}_{i,m',\mathbf{B}_{k'}}} \tag{13}$$

Marginal probabilities are derived from joint probability, and the MI is:

$$I_i(M;\mathbf{B}) = \sum_m\sum_k P_i(m,\mathbf{B}_k)\log\frac{P_i(m,\mathbf{B}_k)}{P_i(m)\cdot P_i(\mathbf{B}_k)} \tag{14}$$

The loss over all layers encourages specialization:

$$\mathcal{L}_{\text{MI}} = -\sum_l\frac{1}{N_{\text{batch}}}\sum_{i=1}^{N_{\text{batch}}} I_i(M;\mathbf{B}) \tag{15}$$

Operating at the bin level (rather than expert level) naturally aligns with device placement granularity in EP: bins with distinct modality affinities can be co-located on devices, reducing cross-device communication.

3.5. Training Objective and Implementation Details

In EP deployments, balanced load within each device is essential. We adopt bin-level load-balancing loss:

$$\mathcal{L}_{\text{bal}} = \sum_l\sum_{k=1}^{N_{\text{bins}}} N_B\sum_{e\in\mathbf{B}_k} f_e\,P_e \tag{16}$$

where f_e and P_e are computed for tokens in bin $\mathbf{B}_k$ (as defined in Sec. 3.1).

The full objective combines task loss, per-bin load-balancing, and inter-bin MI:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha_{\text{bal}}\mathcal{L}_{\text{bal}} + \alpha_{\text{MI}}\mathcal{L}_{\text{MI}} \tag{17}$$

where $\mathcal{L}_{\text{task}}$ is language modeling loss.

We train on 8 NVIDIA A800 GPUs with $N_{\text{bins}} = 8$. For Gaussian-soft modality scores, the temperature is $\tau = 0.5\cdot D$. EMA decay is $\beta = 0.99$ for both Gaussian updates and momentum-adaptive binning. Loss weights are $\alpha_{\text{bal}} = 0.001$ and $\alpha_{\text{MI}} = 0.0001$. Further implementation details and ablation studies are in supplementary material.

4. Results

We conduct extensive experiments to evaluate SMoES across multiple dimensions: effectiveness on diverse VLM and language-only tasks, ablation studies to validate each design choice, visualization of modality specialization, and computational efficiency under expert-parallel deployment.

4.1. Experimental Setup

Model Setup. We build VLMs with CLIP ViT-L/14 vision encoder [44], a 2-layer MLP projector, and MoE language backbones. We evaluate on four architectures spanning different scales and designs: DeepSeekMoE [11], OLMoE [41], Moonlight-MoE [36], and Qwen3-MoE [62]. Models are initialized from public checkpoints and fine-tuned with the same protocol. We compare against four routing strategies: Soft Routing [42], Hard Routing [39], MoIIE [53] (hybrid-routing), and SMAR [61] (KL-divergence).

Training Data and Benchmarks. We follow LLaVA's two-stage training protocol using LLaVA-Pretrain-558K and LLaVA-Instruct-665K datasets respectively [35]. Evaluation covers 10 multimodal benchmarks: MMMU-val/test [64], GQA [23], POPE [28], SQA-img [38], TextVQA [49], MME [16], MMBench/MMBench-CN [37], VQA-v2 [17]; and 6 language benchmarks: MMLU [21], HellaSwag [65], ARC-C/E [9], GSM8k [10], TruthfulQA [32].

Modality Specialization Index (MSI). We introduce MSI to quantify modality-based expert specialization. Modality affiliation probability for expert e at layer l:

$$\tilde{\mathbf{C}}^{(l)}_{m,e} = \frac{\mathbf{C}^{(l)}_{m,e}/\sum_{e'}\mathbf{C}^{(l)}_{m,e'}}{\sum_{m'}\left(\mathbf{C}^{(l)}_{m',e}/\sum_{e'}\mathbf{C}^{(l)}_{m',e'}\right)} \tag{18}$$

MSI measures average deviation from uniform distribution across experts and layers:

$$\text{MSI} = \frac{1}{L}\sum_{l=1}^{L}\frac{1}{E}\sum_{e=1}^{E} 2\cdot\left|\tilde{\mathbf{C}}^{(l)}_{\text{text},e} - 0.5\right| \tag{19}$$

where MSI $\in[0,1]$: 0 = no specialization, 1 = perfect specialization.

4.2. Main Results

We present comprehensive results across all four backbones and 16 benchmarks. Results on DeepSeekMoE and OLMoE are shown in Tab. 1. Results on Moonlight-MoE and Qwen3-MoE are shown in supplementary material.

Across four backbones, SMoES improves over soft routing baseline by 2.2% average (0.9% multimodal, 4.2% language-only). Soft routing lacks modality-specialization guidance and suffers from data imbalance and load-balancing constraints that constrain expert capability.

Table 1. Multimodal and language-only results on VLMs based on DeepSeekMoE and OLMoE. **Bold** and underline: first and second best performance. MSI: Modality Specialization Index. †t/v/s: number of text/vision/shared experts. *[a, b]: KL divergence threshold range.

Method	MSI	Multimodal Tasks (10)											Language-Only Tasks (6)							Overall
		MMMU^{val}	*MMMU^{test}*	*GQA*	*POPE*	*SQA-IMG*	*TextVQA*	*MME*	*MMB*	*MMB-CN*	*VQAv2*	**Avg**	*MMLU*	*HellaSwag*	*ARC-C*	*ARC-E*	*GSM8k*	*TruthfulQA*	**Avg**	**Avg**
VLM based on DeepSeek-MoE (A3B/16B, top-6/64 experts)																				
No Specialization	.177	31.9	30.5	59.5	84.9	68.0	56.5	1718	62.1	61.7	77.4	100%	46.2	49.1	52.3	75.4	10.5	43.8	100%	100%
Hard Routing [39]																				
t32-v32†	1.	30.2	29.2	58.2	**85.6**	63.3	56.3	1555	59.9	57.6	76.2	-3.9%	37.4	42.4	40.4	60.7	2.5	41.2	-26.2%	-12.3%
t48-v16†	1.	30.3	29.7	58.6	84.6	66.0	55.4	1657	62.5	61.5	76.9	-1.8%	41.9	46.6	50.2	72.4	5.3	37.5	-14.5%	-6.6%
MoIIE [53]																				
t16-v16-s32†	.504	32.1	29.9	58.4	85.2	65.1	55.9	1656	62.2	60.1	76.9	-1.5%	41.4	45.6	49.6	72.2	5.8	40.9	-13.1%	-5.8%
t24-v24-s16†	.752	31.9	29.8	58.5	84.6	63.0	55.9	1594	61.1	59.9	76.6	-2.6%	41.2	42.8	47.9	68.2	2.1	42.8	-20.7%	-9.3%
t32-v16-s16†	.800	30.4	29.7	58.5	84.8	64.7	55.8	1667	62.3	61.1	76.9	-1.9%	42.3	50.0	50.9	72.2	6.7	40.2	-9.6%	-4.8%
SMAR [61]																				
d_{KL}*-[0.5, 1.0]**	.543	31.2	**31.5**	58.7	85.5	69.3	57.3	1714	63.6	61.7	77.3	+0.6%	42.1	43.1	47.3	67.3	7.4	45.2	-11.3%	-3.9%
d_{KL}*-[1.5, 2.0]**	.648	32.9	30.2	59.1	85.2	68.3	57.6	**1730**	60.9	58.9	77.3	-0.2%	41.7	44.3	40.6	61.7	6.3	47.7	-15.3%	-5.8%
d_{KL}*-[2.5, 3.0]**	.743	32.2	29.9	58.3	84.9	66.8	57.1	1678	60.6	59.1	77.1	-1.3%	40.7	39.2	32.3	47.5	5.5	46.5	-25.0%	-10.2%
SMoES (ours)																				
attention-soft	.487	**34.7**	31.0	59.6	85.1	69.0	**58.3**	1706	63.5	62.3	77.4	**+1.8%**	**46.5**	**56.4**	**56.0**	**79.0**	**10.9**	46.6	**+6.2%**	**+3.5%**
gaussian-soft	.440	32.4	30.8	**59.9**	85.6	**69.6**	57.5	1689	**65.1**	**62.5**	**77.5**	+1.3%	46.4	53.6	53.9	78.0	10.2	**49.0**	+4.2%	+2.4%
VLM based on OLMoE (A1B/7B, top-8/64 experts)																				
No Specialization	.205	30.0	29.4	58.0	84.9	66.8	**56.6**	**1667**	**62.4**	49.4	75.7	100%	49.9	48.3	59.0	79.1	**32.6**	44.2	100%	100%
Hard Routing [39]																				
t32-v32†	1.	28.4	27.4	57.2	84.8	61.2	50.2	1595	57.3	42.5	74.6	-6.1%	36.3	35.4	40.1	59.6	3.5	42.1	-34.1%	-16.6%
t48-v16†	1.	31.3	29.0	58.2	85.6	65.3	53.3	1603	59.7	45.5	75.6	-2.0%	43.7	42.4	51.4	73.8	13.4	47.0	-16.2%	-7.3%
MoIIE [53]																				
t16-v16-s32†	.509	28.9	28.9	57.6	**85.7**	65.5	53.6	1542	59.4	47.2	75.1	-3.0%	43.6	44.7	51.7	74.1	14.1	47.5	-14.7%	-7.4%
t24-v24-s16†	.754	31.0	28.8	57.6	85.2	62.7	52.3	1507	57.5	44.2	75.0	-4.2%	41.4	44.9	45.8	71.1	3.8	46.2	-23.4%	-11.4%
t32-v16-s16†	.800	32.1	29.3	57.4	85.0	63.8	53.0	1586	60.2	47.5	75.2	-1.8%	41.8	37.2	49.6	74.7	13.9	46.8	-18.7%	-8.2%
SMAR [61]																				
d_{KL}*-[0.5, 1.0]**	.381	32.9	29.6	58.0	84.8	67.6	54.5	1640	60.7	46.3	75.8	-0.3%	48.1	50.8	58.5	77.8	25.0	43.1	-4.5%	-1.9%
d_{KL}*-[1.5, 2.0]**	.485	**33.1**	29.6	57.8	84.8	65.5	54.0	1620	61.5	47.7	75.7	-0.4%	49.4	55.1	58.7	80.2	26.5	46.0	-0.1%	-0.3%
d_{KL}*-[2.5, 3.0]**	.645	32.4	**29.8**	58.0	84.3	66.2	55.6	1650	60.6	47.8	75.8	-0.1%	48.0	50.5	56.5	79.6	26.6	**50.9**	-1.1%	-0.5%
SMoES (ours)																				
attention-soft	.620	31.5	29.7	**58.3**	84.7	66.5	55.8	1644	62.3	50.5	**75.9**	+0.5%	**50.8**	**62.2**	**60.3**	80.8	31.8	47.7	**+6.7%**	**+2.9%**
gaussian-soft	.754	31.4	29.6	58.1	85.1	**67.9**	55.4	1643	62.2	**50.6**	75.7	**+0.6%**	50.1	55.6	58.7	**81.1**	31.4	49.5	+4.3%	+2.0%

Hand-crafted modality specialization (hard/hybrid routing) appears intuitive but inevitably mismatches the complex expert capacity and dynamic data distributions across layers. Hard routing achieves near-perfect modality specialization but incurs substantial performance degradation (-4.4% multimodal, -22.2% language-only). Hybrid routing (MoIIE) provides some recovery (-3.1% multimodal, -17.4% language-only) but remains well below the soft routing baseline. This reveals that rigid specialization cannot be blindly enforced; it must be learned and adaptively tailored to actual expert capacity and data dynamics.

Why do mainstream MoE-VLMs avoid explicit modality control? Our hard/hybrid routing results explain this: incorrectly-designed modality specialization schemes incur severe performance penalties. SMAR introduces an automatic approach through KL divergence regularization rather than explicit hand-crafted assignment. Since the original SMAR was designed for models with small expert counts (no more than eight), while ours has numerous small experts, we test various KL divergence regularization strengths in our setting. SMAR achieves significant MSI gain while maintaining more competitive performance: incurring only 1.0% and 15.1% performance loss on multimodal and language-only tasks. However, SMAR suffers from incompatibility between KL divergence regularization and load-balancing constraints (it disabled load-balancing in the final model), making it difficult to leverage specialization in MoE models with numerous small experts.

Our SMoES addresses this by unifying soft modality scores and inter-bin MI objectives, achieving strong modality specialization while simultaneously maximizing expert capacity utilization and maintaining load balance. Our attention-based soft modality estimation achieves average gains of 1.0% on multimodal tasks and 4.4% on language-only tasks, while our Gaussian-statistics variant achieves 0.9% and 4.1% respectively. Both significantly outperform

Table 2. Ablation on the modality score type (DeepSeekMoE).

Method	MSI	Multi-Modal	Language	Overall
No Specialization	.177	100%	100%	100%
SMAR (best)	.543	+0.6%	-11.3%	-3.9%
SMoES				
hard-score	.904	-0.8%	+0.5%	-0.3%
attention-soft	.487	**+1.8%**	**+6.2%**	**+3.5%**
gaussian-soft	.440	+1.3%	+4.2%	+2.4%

Table 3. Ablation on inter-bin specialization (DeepSeekMoE).

Method	MSI	Multi-Modal	Language	Overall
No Specialization	.177	100%	100%	100%
w/ binning	.415	+0.9%	+3.0%	+1.7%
w/ inter-bin				
KL	.724	-1.5%	-8.5%	-4.1%
MI-attention	.487	**+1.8%**	**+6.2%**	**+3.5%**
MI-gaussian	.440	+1.3%	+4.2%	+2.4%

all baseline methods, validating that well-designed learnable specialization schemes can unlock performance gains in large-scale MoE-VLMs while preserving load balance.

4.3. Ablation Studies

4.3.1. Modality Score and Specialization Objective

We first demonstrate the effectiveness of soft modality scores. Hard-score uses binary labels based on input source modality; although it achieves high MSI, it cannot improve model performance. Both attention-soft and gaussian-soft scores significantly outperform hard-score (Tab. 2). Notably, hard-score with MI-based specialization still exceeds the best SMAR configuration, validating the importance of our specialization objective.

We next evaluate our MI-based specialization with expert binning (Tab. 3). Expert-binning itself improves performance (+1.7%) by providing structure for modality-aware specialization. Our inter-bin MI objective further boosts gains (+3.5% for attention-soft, +2.4% for gaussian-soft), demonstrating that MI effectively guides experts toward coherent modality-specific patterns. In contrast, KL-divergence in SMAR fails to improve performance.

4.3.2. Expert Binning Strategy and Granularity

We evaluate our adaptive binning strategy (based on modality-aware EMA statistics) versus fixed binning (original expert order). Table Tab. 4 shows adaptive binning consistently outperforms fixed binning across both binning-only and MI specialization settings, confirming the effectiveness of modality-aware bin formation.

We next examine bin granularity by varying the number of bins (Fig. 5). Too many bins can increase deployment imbalance despite enabling finer specialization, while too few bins reduce specialization effectiveness.

Table 4. Ablation on expert binning strategies (DeepSeekMoE).

Method	MSI	Multi-Modal	Language	Overall
No Specialization	.177	100%	100%	100%
w/ binning				
fixed	.357	+0.9%	+2.9%	+1.6%
adaptive	.415	+0.9%	+3.0%	+1.7%
attention-soft				
fixed	.450	+2.0%	+0.2%	+1.3%
adaptive	.487	+1.8%	+6.2%	+3.5%
gaussian-soft				
fixed	.398	+1.9%	-1.0%	+0.8%
adaptive	.440	+1.3%	+4.2%	+2.4%

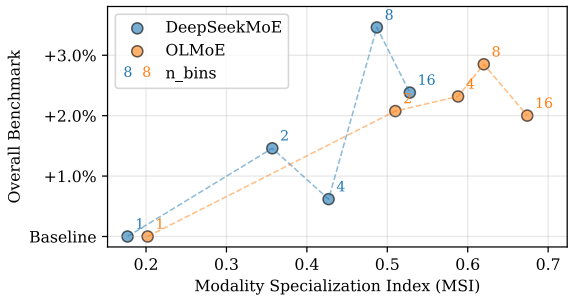

Figure 5. Ablation on number of expert bins ($\text{SMoES}_{\text{attention-soft}}$).

4.4. Visualization and Analysis

SMoES achieves clear bin-modality correspondences, as visualized in Fig. 6. Early layers exhibit higher MSI and sharper expert-modality separation, while deeper layers show balanced distributions with more modality fusion.

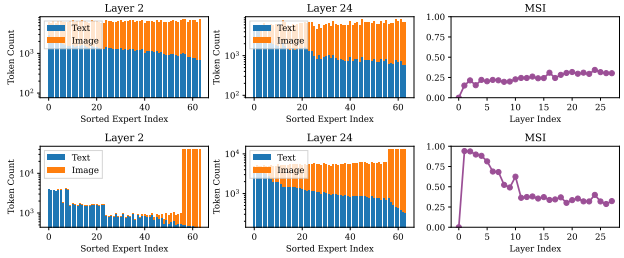

Figure 6. Routing distribution of tokens to experts in DeepSeek-MoE. First row: soft baseline; second row: $\text{SMoES}_{\text{attention-soft}}$.

Expert modality specialization evolves dynamically during training, as shown in Fig. 7. Soft routing without specialization results in experts handling both modalities simultaneously, constraining capacity. Our method shows sharper differentiation in shallow layers where tokens maintain clearer modality identity, and more fusion-aware adaptation in deeper layers.

Our soft modality scores exhibit smooth fusion transitions across layers, as illustrated in Fig. 8. Both attention-soft and Gaussian-soft estimators show increasing fusion intensity toward deeper layers. A slight difference is that attention-soft is biased toward text, while Gaussian-soft is

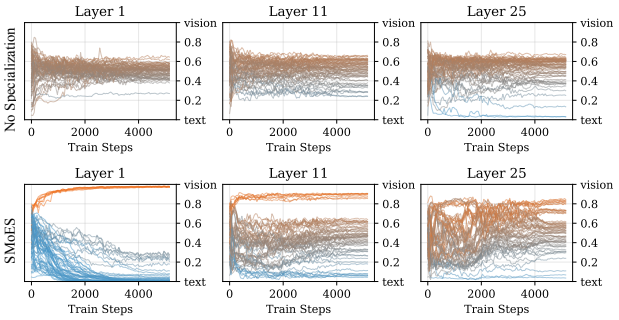

Figure 7. Evolution of expert specialization during training.

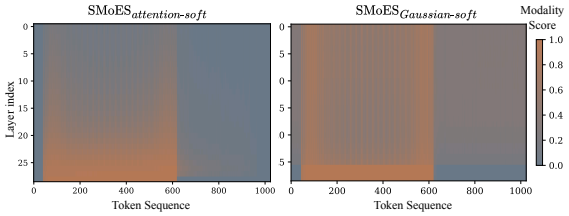

Figure 8. Soft modality score across layers in DeepSeekMoE.

biased toward vision in high-fusion regimes.

4.5. Efficiency Analysis

Beyond task accuracy, SMoES reduces expert-parallel (EP) communication overhead by aligning expert placement with modality preferences. We deploy on two NVIDIA Orin GPUs via 10Gb Ethernet using EP, representing a typical edge-side scenario in autonomous vehicles (Fig. 9). EP avoids weight and KV Cache redundancy compared to TP/DP, making it memory-efficient for edge resources. Baseline uses synchronous transmission since balanced experts provide no benefit from asynchrony, while SMoES's increased local expert concentration enables asynchronous transmission, overlapping communication and computation similar to PD separation.

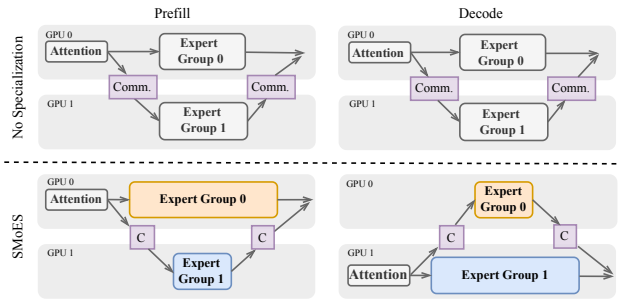

Figure 9. Expert-parallel (EP) deployment on two GPUs.

Tab. 5 shows cross-GPU EP transfer ratios for vision and text tokens at prefill and decode stages on OLMoE. Expert specialization routes tokens to local experts more frequently, reducing inter-device communication. We report separate ratios for vision and text tokens (quantities differ between phases) and the overall V+T ratio. DeDup (de-duplication) is employed to avoid duplicate transmission of tokens routed to experts on the same device in top-k routing.

Table 5. Cross-GPU EP transfer ratio at prefill/decode stages. V: Vision tokens; T: Text tokens.

Benchmark	Method	Prefill (P) V	Prefill (P) T	Prefill (P) V+T	Decode (D) T
MMMU (PV:PT:DT=32:8:1)	baseline	97.7%	99.5%	98.0%	86.5%
	SMoES	15.0%	99.3%	31.1%	43.3%
	Δ	↓84.6%	↓0.3%	↓68.3%	↓49.9%
SQA-IMG (PV:PT:DT=14:7:1)	baseline	97.5%	99.7%	98.2%	94.9%
	SMoES	13.0%	99.2%	40.6%	49.9%
	Δ	↓86.6%	↓0.5%	↓58.7%	↓47.4%

Table 6. TTFT and TPOT speed improvement of SMoES compared to soft-routing baseline. Δ: speedup percentage.

Benchmark	Method	Batch Size=1 TTFT(s)	Batch Size=1 TPOT(s)	Batch Size=8 TTFT(s)	Batch Size=8 TPOT(s)
MMMU	baseline	2.810	0.786	7.949	1.414
	SMoES	**2.519**	**0.703**	**6.203**	**1.287**
	Δ	↓10.3%	↓10.5%	↓22.0%	↓9.0%
SQA-IMG	baseline	1.493	0.766	5.824	1.278
	SMoES	**1.356**	**0.692**	**4.859**	**1.134**
	Δ	↓9.2%	↓9.7%	↓16.6%	↓11.3%

Tab. 6 shows TTFT for Prefill and TPOT for Decode. Performance improvement stems from reduced communication overhead and parallel execution of computation and communication. Larger batch sizes in Prefill increase communication proportion, yielding greater gains, while Decode maintains stable improvement ratios across batch sizes due to fewer activated experts.

5. Conclusion

In this paper, we addressed the challenge of modality-guided expert specialization in MoE-VLMs. We introduced SMoES, which consists of dynamic soft modality scores that capture layer-dependent fusion patterns, an expert binning mechanism aligned with expert-parallel deployment, and an inter-bin mutual information regularization that encourages coherent modality specialization. Extensive experiments across four MoE backbones and 16 benchmarks validate our approach, demonstrating consistent improvements in task accuracy while simultaneously reducing communication overhead and increasing throughput in expert-parallel deployments. Our work has some limitations, for example, the Gaussian-soft estimator currently uses a simple unimodal Gaussian with diagonal covariance for efficiency. Although we have explored GMM as a preliminary extension in the supplementary material, further refining richer density models remains an open problem. This work demonstrates the value of data-driven modality specialization and opens promising avenues for future exploration in expert specialization for MoE-based multimodal learning.

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Soft Modality-Guided Expert Specialization in MoE-VLMs

Supplementary Material

1. Detailed Experimental Setup

1.1. Vision Encoder and Projector

All VLM configurations use the same vision encoder and projector so that backbone and routing changes are the only variables:

Vision Encoder: CLIP ViT-L/14 [44] with 336×336 input resolution, producing 576 visual tokens for every image. **Projector:** A two-layer MLP with GELU activation that maps the 1024-dimensional CLIP features to the hidden size of each MoE backbone (e.g., 2048 for DeepSeekMoE).

1.2. MoE Backbone Architectures

We provide detailed specifications for the four MoE-based VLM backbones used in our experiments. They cover different router designs (softmax vs. sigmoid), expert settings (with or without shared experts), attention variants (MHA, MLA, GQA), and model scales from 7B to 30B parameters. This variety allows us to validate SMoES under heterogeneous architectures. Tab. S2 summarizes the main differences.

DeepSeekMoE [11] employs two shared experts together with 64 routed experts under top-6 gating, providing both stable training and sufficient specialization capacity.

OLMoE [41] is an open-source lightweight (only 7B parameters in total) MoE model with 64 experts and top-8 routing. Unlike DeepSeekMoE, it does not use shared experts, relying entirely on the routing mechanism for expert selection. It is not optimized for multilingual tasks, which can be seen in the MMB-CN benchmark.

Moonlight-MoE [36] is a recent architecture that is similar to DeepSeekMoE in scale. However, it differs significantly in its router design: unlike other models that use softmax-based routing, Moonlight-MoE employs sigmoid-based routing with router score bias, which provides a different gating mechanism for expert selection. The introduction of router score bias also induces certain fluctuations to the router, for example, hard routing does not perfectly separate visual and text tokens (see MSI in Tab. S4).

Qwen3-MoE [62] is a 30B-parameter production model with 128 experts and top-8 routing. It uses GQA attention and head-wise QK norm, and serves as the strongest baseline among the four backbones.

1.3. Training Configuration

We follow the standard LLaVA [35] two-stage training protocol:

Stage 1: Feature Alignment. We train only the projector while keeping the vision encoder and language model frozen. This stage uses 558K image-caption pairs from the LLaVA-Pretrain dataset.

Stage 2: Instruction Tuning. We fine-tune the entire model (except the vision encoder) on the LLaVA-Instruct-665K dataset with diverse vision-language instructions.

Table S1. Training details.

	Stage 1	Stage 2
Optimizer	AdamW	AdamW
$\text{LR}_{\text{vision-encoder}}$	0	0
$\text{LR}_{\text{projector}}$	1e-3	2e-5
$\text{LR}_{\text{language-model}}$	0	2e-5
LR Scheduler	cosine	cosine
Warmup Ratio	0.03	0.03
Num Samples	558K	665K
Num Epochs	1	1
Batch Size	256	128
Weight Decay	0.0	0.0
Gradient Clipping	1.0	1.0

We show training details in Tab. S1. All experiments use AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay 0.0. For our method-specific hyperparameters, we set $N_{\text{bins}} = 8$, $\alpha_{\text{bal}} = 0.001$, $\alpha_{\text{MI}} = 0.0001$, EMA decay $\beta = 0.99$, and Gaussian temperature $\tau = 0.5 \cdot D$ as default values across all backbones.

All experiments are conducted on 8 NVIDIA A800-80GB GPUs using mixed-precision training (BF16) and gradient checkpointing to reduce memory usage. In cases where memory constraints persist, we use gradient accumulation to achieve the effective batch sizes. During benchmark evaluation, we use greedy search without sampling to produce consistent results. We set the maximum number of new tokens in generation to 256 for the GSM8k benchmark and 20 for the others.

We do not use dynamic expert migration during training in our experiments. However, a reasonable migration frequency according to the expert specialization is encouraged when scaling. Thanks to EMA, expert groupings remain stable over a certain period, making the migration overhead reasonable.

1.4. Efficient Implementation of SMoES

Although the attention-accumulated score formulation (Sec. 3.2.1) in SMoES uses the attention matrix $\text{Attn}^{(l)}$ for clarity, the aggregation can be efficiently implemented without materializing the full attention matrix, making it compatible with memory-efficient attention kernels such

Table S2. Detailed specifications of MoE backbone architectures.

Architecture	DeepSeekMoE	OLMoE	Moonlight-MoE	Qwen3-MoE
Huggingface Checkpoint	deepseek-moe-16b-base	OLMoE-1B-7B-0125	Moonlight-16B-A3B	Qwen3-30B-A3B
Total Parameters	16B	7B	16B	30B
Activated Params	3B	1B	3B	3B
Number of Routed Experts	64	64	64	128
Top-k Routing	6	8	6	8
Number of Shared Experts	2	0	2	0
Router Type	softmax	softmax	sigmoid	softmax
Router Score Bias	no	no	yes	no
Hidden Dimension	2048	2048	2048	2048
Number of Layers	28	16	27	48
Dense Replace for First k Layers	1	0	1	0
QK Norm	no	token-wise	no	head-wise
Attention Type	MHA	MHA	MLA	GQA

as Flash Attention [12]. Specifically, we concatenate the modality scores $M_{ij,m}^{\text{attn},(l)}$ to the value vectors before applying attention, allowing the kernel to perform the weighted aggregation implicitly. After the attention operation, we extract the aggregated modality scores from the corresponding dimensions of the attention output. This implementation strategy preserves the aggregation semantics while leveraging efficient attention implementations without additional memory or computational overhead.

2. Supplementary Results

2.1. Main results on Moonlight-MoE and Qwen3-MoE

In addition to the primary evaluation on DeepSeekMoE and OLMoE reported in Tab. 1, we provide the detailed performance on Moonlight-MoE and Qwen3-MoE in Tab. S4. The results demonstrate that SMoES consistently outperforms various routing baselines across these backbones, exhibiting a performance gain trend similar to that observed on DeepSeekMoE and OLMoE. For Moonlight-MoE, we notice that the improvement in the Modality Specialization Index (MSI) is relatively moderate, increasing to 0.442 from 0.380 in the vanilla soft routing baseline. This phenomenon likely stems from the architectural characteristics and pretraining strategies of Moonlight-MoE, which appear to inherently foster a certain level of modality differentiation even without explicit routing constraints. Despite this existing bias, SMoES still effectively refines the routing mechanism to achieve a more structured expert specialization.

Beyond the multimodal performance, we observe a critical limitation in existing modality-specialized methods: they often suffer from substantial performance degradation on language-only tasks. Such a decline suggests that traditional rigid routing or aggressive specialization constraints can inadvertently compromise the model's fundamental linguistic knowledge. While the instruction-tuning data incorporates pure text samples, these baseline methods

Table S3. Ablation on the modality score type in SMoES.

Method	MSI	Multi-Modal	Language	Overall
VLM based on DeepSeekMoE (A3B/16B, top-6/64 experts)				
No Specialization	.177	100%	100%	100%
SMAR (best)	.543	+0.6%	-11.3%	-3.9%
SMoES				
hard-score	.904	-0.8%	+0.5%	-0.3%
attention-soft	.487	**+1.8%**	**+6.2%**	**+3.5%**
gaussian-soft	.440	+1.3%	+4.2%	+2.4%
VLM based on OLMoE (A1B/7B, top-8/64 experts)				
No Specialization	.205	100%	100%	100%
SMAR (best)	.485	-0.4%	-0.1%	-0.3%
SMoES				
hard-score	.756	-0.0%	+1.9%	+0.7%
attention-soft	.620	+0.5%	**+6.7%**	**+2.9%**
gaussian-soft	.754	**+0.6%**	+4.3%	+2.0%
VLM based on Moonlight-MoE (A3B/16B, top-6/64 experts)				
No Specialization	.380	100%	100%	100%
SMAR (best)	.673	-2.8%	-30.3%	-13.1%
SMoES				
hard-score	.451	+0.1%	+3.1%	+1.2%
attention-soft	.442	+0.2%	+4.5%	+1.8%
gaussian-soft	.449	**+0.7%**	**+8.0%**	**+3.5%**
VLM based on Qwen3-MoE (A3B/30B, top-8/128 experts)				
No Specialization	.284	100%	100%	100%
SMAR (best)	.553	+0.7%	-5.4%	-1.6%
SMoES				
hard-score	.890	-1.8%	-1.5%	-1.7%
attention-soft	.726	**+1.3%**	**+0.1%**	**+0.9%**
gaussian-soft	.566	+0.8%	-0.2%	+0.4%

Table S4. Multimodal and language-only results on VLMs based on Moonlight-MoE and Qwen3-MoE. **Bold** and underline indicate the first and second best performance. MSI denotes Modality Specialization Index. †t/v/s denote the number of text/vision/shared experts. *[a, b] denotes KL divergence threshold range.

Method	MSI	Multimodal Tasks (10)											Language-Only Tasks (6)							Overall
		MMMU^{val}	*MMMU^{test}*	*GQA*	*POPE*	*SQA-IMG*	*TextVQA*	*MME*	*MMB*	*MMB-CN*	*VQAv2*	**Avg**	*MMLU*	*HellaSwag*	*ARC-C*	*ARC-E*	*GSM8k*	*TruthfulQA*	**Avg**	**Avg**
VLM based on Moonlight-MoE (A3B/16B, top-6/64 experts)																				
No Specialization	.380	34.1	31.0	58.8	85.7	67.7	58.9	**1715**	67.0	63.7	77.0	100%	51.0	54.1	60.1	78.3	25.3	**51.4**	100%	100%
Hard Routing [39]																				
t32-v32†	.991	27.1	27.6	57.9	85.0	59.9	54.4	1525	59.9	59.8	76.1	-8.2%	33.1	39.5	40.1	53.7	3.4	43.1	-38.3%	-19.5%
t48-v16†	.952	31.2	29.6	58.5	85.4	64.1	58.1	1597	64.8	60.0	76.6	-3.7%	41.8	47.3	49.5	67.9	14.1	44.4	-19.9%	-9.8%
MoIIE [53]																				
t16-v16-s32†	.686	32.2	29.9	58.4	85.7	65.5	58.2	1636	63.8	61.3	76.7	-2.8%	44.1	45.6	53.2	72.9	13.1	46.3	-17.7%	-8.3%
t24-v24-s16†	.836	29.8	28.4	58.4	85.7	63.4	56.8	1615	61.7	59.8	76.5	-5.2%	35.4	39.6	39.8	53.8	4.9	42.1	-36.9%	-17.1%
t32-v16-s16†	.833	32.4	29.8	58.6	**85.7**	65.1	57.4	1583	63.5	60.6	76.6	-3.4%	43.7	47.5	52.8	70.3	12.4	40.3	-20.3%	-9.7%
SMAR [61]																				
d_{KL}*-[0.5, 1.0]**	.599	32.3	30.3	58.3	84.1	66.9	57.5	1590	65.7	61.3	76.7	-2.7%	38.9	38.1	47.0	64.3	11.7	32.6	-30.6%	-13.2%
d_{KL}*-[1.5, 2.0]**	.646	31.6	29.3	58.2	84.6	66.1	55.8	1626	64.7	60.8	76.2	-3.7%	36.8	39.1	39.9	55.4	6.9	26.6	-39.9%	-17.3%
d_{KL}*-[2.5, 3.0]**	.673	31.9	29.6	58.1	84.9	68.1	57.7	1692	63.7	59.7	76.4	-2.8%	40.0	37.2	44.6	62.8	11.1	37.6	-30.3%	-13.1%
SMoES (ours)																				
attention-soft	.442	33.7	**32.2**	58.8	85.1	67.4	58.5	1668	**68.2**	**64.7**	77.0	+0.2%	**52.9**	55.6	67.7	84.1	26.4	49.6	+4.5%	+1.8%
gaussian-soft	.449	**35.7**	32.0	**59.0**	85.2	**68.6**	**58.9**	1671	67.0	64.1	**77.0**	**+0.7%**	52.9	**56.3**	**68.9**	**85.0**	**30.8**	49.2	**+8.0%**	**+3.5%**
VLM based on Qwen3-MoE (A3B/30B, top-8/128 experts)																				
No Specialization	.284	39.8	37.4	57.3	85.7	76.5	58.6	**1830**	67.9	**71.7**	75.6	**100%**	**67.9**	82.2	87.8	94.5	66.1	60.7	100%	100%
Hard Routing [39]																				
t64-v64†	1.	37.0	31.9	56.1	85.4	66.2	55.4	1630	66.4	60.0	74.5	-7.4%	51.9	68.5	75.2	87.2	34.2	54.8	-20.1%	-12.1%
t96-v32†	1.	39.6	34.6	58.0	86.6	71.7	59.1	1741	68.7	66.5	75.1	-2.3%	60.5	77.9	81.3	91.1	53.3	58.8	-8.3%	-4.5%
MoIIE [53]																				
t32-v32-s64†	.509	39.9	34.6	57.0	85.9	70.8	57.7	1766	69.2	66.7	75.2	-2.5%	60.9	79.3	78.7	87.3	52.6	58.2	-9.4%	-5.1%
t48-v48-s32†	.754	37.8	33.5	56.9	85.6	70.7	56.8	1649	68.7	62.0	74.9	-5.0%	55.9	73.0	71.1	84.6	46.0	53.0	-16.9%	-9.5%
t64-v32-s32†	.800	38.3	34.5	57.4	85.6	72.1	58.6	1693	67.0	65.3	74.7	-3.6%	60.9	78.1	83.3	91.9	54.1	59.7	-7.2%	-5.0%
SMAR [61]																				
d_{KL}*-[0.5, 1.0]**	.747	**44.8**	38.6	56.8	**86.8**	72.9	57.9	1711	60.6	63.7	75.2	-1.9%	56.5	69.9	75.9	86.0	56.4	**63.4**	-10.8%	-5.2%
d_{KL}*-[1.5, 2.0]**	.553	43.5	**40.0**	56.3	85.5	**78.2**	58.8	1664	69.9	69.4	75.3	+0.7%	60.6	74.3	82.7	92.2	61.1	63.0	-5.4%	-1.6%
d_{KL}*-[2.5, 3.0]**	.647	41.9	38.7	56.2	86.1	77.6	59.8	1718	69.7	69.3	75.3	+0.4%	60.8	74.5	81.1	91.1	60.3	59.7	-6.9%	-2.4%
SMoES (ours)																				
attention-soft	.726	40.5	38.9	58.2	85.5	75.5	**60.8**	1775	**72.7**	71.0	75.9	**+1.3%**	67.7	**83.0**	88.0	96.1	**66.2**	59.5	**+0.1%**	**+0.9%**
gaussian-soft	.566	41.7	37.9	**58.9**	85.8	76.0	59.2	1760	71.5	69.3	**76.1**	+0.8%	67.5	82.7	**88.4**	**96.7**	65.2	59.0	-0.2%	+0.4%

tend to erode the model's ability to handle complex linguistic tasks, particularly those requiring multi-step reasoning like GSM8k. In the broader context of VLM development, language capabilities are frequently undervalued; however, they remain indispensable for ensuring a high-quality user experience and enabling sophisticated long-context reasoning. Our results underscore that SMoES successfully maintains robust language performance while enhancing multimodal synergy, which is vital for building truly unified and versatile multimodal models.

2.2. Gaussian Estimator

In our primary methodology, we employ a unimodal Gaussian Estimator with a diagonal covariance matrix, primarily to ensure computational efficiency and numerical stability in high-dimensional feature spaces. However, such a simplified distribution may be overly restrictive, potentially failing to capture the intricate, non-linear dependencies and multi-modal structures inherent in fused modality representations. To investigate the impact of estimator expressiveness, we conduct preliminary experiments exploring more sophisticated density modeling techniques. While adopting a full covariance matrix could theoretically cap-

Benchmark	Baseline	k=1	k=2	k=4
multimodal	100%	+0.6%	**+1.2%**	+0.8%
language	100%	+4.3%	**+5.6%**	+4.3%
Overall	100%	+2.0%	**+2.8%**	+2.1%

Table S5. Gaussian Mixture Model (GMM) estimator with different n-components on OLMoE.

Table S6. Ablation on inter-bin specialization objectives. KL: Kullback-Leibler divergence. MI: Mutual Information.

Method	MSI	Multi-Modal	Language	Overall
VLM based on DeepSeekMoE (A3B/16B, top-6/64 experts)				
No Specialization	.177	100%	100%	100%
w/ binning	.415	+0.9%	+3.0%	+1.7%
w/ Inter-bin				
KL	.724	-1.5%	-8.5%	-4.1%
MI-attention	.487	**+1.8%**	**+6.2%**	**+3.5%**
MI-gaussian	.440	+1.3%	+4.2%	+2.4%
VLM based on OLMoE (A1B/7B, top-8/64 experts)				
No Specialization	.205	100%	100%	100%
w/ binning	.324	-12.2%	-9.6%	-11.2%
w/ Inter-bin				
KL	.545	+0.4%	-0.9%	-0.1%
MI-attention	.620	+0.5%	**+6.7%**	**+2.9%**
MI-gaussian	.754	**+0.6%**	+4.3%	+2.0%
VLM based on Moonlight-MoE (A3B/16B, top-6/64 experts)				
No Specialization	.380	100%	100%	100%
w/ binning	.441	-0.0%	-1.5%	-0.6%
w/ Inter-bin				
KL	.748	-5.9%	-8.1%	-6.7%
MI-attention	.442	+0.2%	+4.5%	+1.8%
MI-gaussian	.449	**+0.7%**	**+8.0%**	**+3.5%**
VLM based on Qwen3-MoE (A3B/30B, top-8/128 experts)				
No Specialization	.284	100%	100%	100%
w/ binning	.468	+0.6%	-0.1%	+0.3%
w/ Inter-bin				
KL	.540	-2.0%	**+1.0%**	-1.0%
MI-attention	.726	**+1.3%**	+0.1%	+0.9%
MI-gaussian	.566	+0.8%	-0.2%	+0.4%

Table S7. Ablation on expert binning strategy.

Method	MSI	Multi-Modal	Language	Overall
VLM based on DeepSeekMoE (A3B/16B, top-6/64 experts)				
No Specialization	.177	100%	100%	100%
w/ binning				
fixed	.357	+0.9%	+2.9%	+1.6%
adaptive	.415	+0.9%	+3.0%	+1.7%
attention-soft				
fixed	.450	+2.0%	+0.2%	+1.3%
adaptive	.487	+1.8%	+6.2%	+3.5%
gaussian-soft				
fixed	.398	+1.9%	-1.0%	+0.8%
adaptive	.440	+1.3%	+4.2%	+2.4%
VLM based on OLMoE (A1B/7B, top-8/64 experts)				
No Specialization	.205	100%	100%	100%
w/ binning				
fixed	.324	-18.6%	-10.5%	-15.6%
adaptive	.324	-12.2%	-9.6%	-11.2%
attention-soft				
fixed	.355	+0.1%	+3.3%	+1.4%
adaptive	.620	+0.5%	+6.7%	+2.9%
gaussian-soft				
fixed	.433	-0.6%	+1.8%	+0.5%
adaptive	.754	+0.6%	+4.3%	+2.0%
VLM based on Moonlight-MoE (A3B/16B, top-6/64 experts)				
No Specialization	.380	100%	100%	100%
w/ binning				
fixed	.445	-9.8%	-37.3%	-20.2%
adaptive	.441	-0.0%	-1.5%	-0.6%
attention-soft				
fixed	.377	+0.1%	-0.6%	-0.2%
adaptive	.442	+0.2%	+4.5%	+1.8%
gaussian-soft				
fixed	.388	+0.3%	+5.0%	+2.1%
adaptive	.449	+0.7%	+8.0%	+3.5%
VLM based on Qwen3-MoE (A3B/30B, top-8/128 experts)				
No Specialization	.284	100%	100%	100%
w/ binning				
fixed	.308	-0.1%	+0.6%	+0.1%
adaptive	.468	+0.6%	-0.1%	+0.3%
attention-soft				
fixed	.325	+1.3%	+1.8%	+1.5%
adaptive	.726	+1.3%	+0.1%	+0.9%
gaussian-soft				
fixed	.332	+0.6%	+0.0%	+0.4%
adaptive	.566	+0.8%	-0.2%	+0.4%

ture cross-dimension correlations, it introduces $O(D^2)$ parameters and incurs significant overhead for online updates and likelihood computations. As a more practical alternative, we evaluate the Gaussian Mixture Model (GMM) with $k \in \{2, 4\}$ components, as detailed in Tab. S5. Our results indicate that $k = 2$ yields measurable performance gains, which aligns with our qualitative observations in Figs. S10 to S12 where certain layers exhibit distinct bi-modal or multi-peak token distributions. These findings suggest that the occasional performance gap between the Gaussian and attention-based estimators likely arises from the limited capacity of a simple Gaussian model to characterize highly complex or heterogeneous token samples.

2.3. Ablation study

Ablation on the modality score type on all four MoE backbones is shown in Tab. S3. Our attention-soft and gaussian-soft scores outperform the hard routing score (0/1 hard-coded) on all four MoE backbones. This demonstrates the importance of soft modality scores in modality differentiation. Furthermore, even the hard modality score can outperform SMAR, indicating that MI-based modality differentiation is more effective than KL-based approaches.

Ablation on inter-bin specialization objective is shown in Tab. S6. Expert binning can increase MSI since it provides the feasibility for modality differentiation. Inter-bin MI-based specialization further enhances modality differen-

Table S8. Ablation on Gaussian temperature. D: number of the feature dimension.

Method	MSI	Multi-Modal	Language	Overall
VLM based on DeepSeekMoE (A3B/16B, top-6/64 experts)				
No Specialization	.177	100%	100%	100%
gaussian-τ				
$0.05D$	.516	-0.9%	**+4.8%**	+1.2%
$0.1D$	.574	+0.6%	+4.2%	+1.9%
$0.3D$	.564	**+1.6%**	+1.4%	+1.5%
$0.5D$ *(default)*	.440	+1.3%	+4.2%	**+2.4%**
$0.7D$	.444	+1.4%	+3.1%	+2.1%
$1.0D$	.475	+0.2%	+2.4%	+1.1%
VLM based on OLMoE (A1B/7B, top-8/64 experts)				
No Specialization	.205	100%	100%	100%
gaussian-τ				
$0.05D$	.744	-0.4%	+0.6%	-0.0%
$0.1D$	.744	-0.1%	**+4.3%**	+1.7%
$0.3D$	.759	-0.4%	+4.3%	+1.4%
$0.5D$ *(default)*	.754	**+0.6%**	+4.3%	**+2.0%**
$0.7D$	.745	+0.2%	+3.9%	+1.6%
$1.0D$	.740	+0.5%	-2.0%	-0.5%

tiation and improves model performance. In contrast, KL-based inter-bin modality differentiation, despite achieving higher modality differentiation degrees on DeepSeekMoE and Moonlight-MoE, degrades model performance, which is unacceptable.

Ablation on expert binning strategy is shown in Tab. S7. We compare our momentum-adaptive binning strategy (adaptive) to the static, predefined binning (fixed) strategy. In all cases, adaptive binning is superior to fixed binning in terms of model performance. Furthermore, in most cases, adaptive binning yields a higher degree of modality specialization. This is because the pretrained MoE backbone already exhibits certain expert capability tendencies, and forcing predefined modality bins for experts would interfere with their capability expression. Furthermore, during training, experts may gradually differentiate, and fixed binning may constrain the experts' modality differentiation capacity.

Ablation on Gaussian temperature in gaussian-soft SMoES is shown in Tab. S8. We evaluate $\tau \in \{0.05D, 0.1D, 0.3D, 0.5D, 1.0D\}$, where D is the feature dimension. As shown in the table, $0.5D$ provides the best balance between confident and smooth modality scores. The table shows that both excessively high and low temperatures lead to performance degradation. Moderate temperature levels yield similar and stable performance improvements. Therefore, we adopt a moderate value of $0.5D$ as the default setting.

Ablation on modality specialization loss weight (α_{MI}) and load balance loss weight (α_{bal}) in attention-soft SMoES is shown in Tab. S9. We evaluate α_{MI} and α_{bal} in $\{1e-1, 1e-2, 1e-3, 1e-4, 1e-5\}$. The table shows that when α_{MI} is relatively large, it is possible to achieve substantial model performance improvements, but it may also cause training instability. Therefore, we choose a relatively low α_{MI} ($1e-4$) to obtain stable performance improvements. For α_{bal}, the results may vary across different models. For example, on DeepSeekMoE, using a smaller balance loss can further improve model performance, while on OLMoE, a moderate balance loss is preferred. To achieve stable performance and ensure the magnitude is consistent with the balance loss of the soft routing baseline, we choose $1e-3$ as the weight for the balance loss.

Table S9. Ablation on loss alpha ($\text{SMoES}_{\text{attention-soft}}$).

Method	MSI	Multi-Modal	Language	Overall
VLM based on DeepSeekMoE (A3B/16B, top-6/64 experts)				
No Specialization	.177	100%	100%	100%
α_{MI}				
1e-1	.533	**+3.0%**	**+16.1%**	**+8.6%**
1e-2	.716	-11.9%	+1.0%	-6.3%
1e-3	.610	-1.0%	+4.9%	+1.5%
1e-4 (default)	.487	+1.8%	+6.2%	+3.5%
1e-5	.417	-0.0%	+4.0%	+1.7%
α_{bal}				
1e-1	.290	-43.2%	-48.4%	-45.3%
1e-2	.410	-7.1%	-17.8%	-11.3%
1e-3 (default)	.487	+1.8%	+6.2%	+3.5%
1e-4	.578	**+3.9%**	**+8.1%**	**+5.2%**
1e-5	.550	+3.9%	+6.0%	+4.5%
VLM based on OLMoE (A1B/7B, top-8/64 experts)				
No Specialization	.205	100%	100%	100%
α_{MI}				
1e-1	.684	+1.8%	+4.6%	+2.8%
1e-2	.871	**+2.0%**	**+8.3%**	**+4.3%**
1e-3	.982	-0.2%	+2.2%	+0.7%
1e-4 (default)	.620	+0.5%	+6.7%	+2.9%
1e-5	.340	+0.3%	+3.9%	+1.7%
α_{bal}				
1e-1	.738	-15.4%	-27.9%	-20.1%
1e-2	.796	-0.3%	-1.8%	-0.9%
1e-3 (default)	.620	**+0.5%**	**+6.7%**	**+2.9%**
1e-4	.654	+0.4%	+4.9%	+2.1%
1e-5	.702	+0.3%	+6.0%	+2.6%

2.4. Efficiency Analysis

To evaluate the practical advantages of our SMoES method in real-world scenarios, we conduct edge-side inference experiments comparing our SMoES against baseline models without modality specialization. These experiments demonstrate the communication efficiency gains achieved through expert specialization, particularly in resource-constrained edge deployment settings.

We deploy our experiments on a dual-NVIDIA-Orin GPU setup, which represents a typical edge-side scenario

Table S10. Cross-GPU EP transfer ratio at prefill and decode stages for OLMoE. V: Vision tokens; T: Text tokens. *(PV : PT : DT)*: token count ratio for prefill-vision, prefill-text and decode-text. DeDup: de-duplication of tokens routed to experts on the same device in top-K routing.

Method	Top-K w/o DeDup				Top-K w/ DeDup			
	Prefill (P)			**Decode (D)**	**Prefill (P)**			**Decode (D)**
	V	**T**	**V+T**	**T**	**V**	**T**	**V+T**	**T**
MMMU (PV : PT : DT = 79% : 19% : 2%)								
Baseline (No Specialization)	42.7%	70.4%	48.0%	37.4%	97.7%	99.5%	98.0%	86.5%
$\text{SMoES}_{\text{attention-soft}}$	8.8% ↓ 79.4%	71.7% ↑ 1.0%	20.7% ↓ 56.9%	34.3% ↓ 8.3%	33.6% ↓ 65.6%	98.9% ↓ 0.6%	46.1% ↓ 53.0%	88.9% ↑ 2.8%
$\text{SMoES}_{\text{gaussian-soft}}$	3.3% ↓ 92.3%	76.7% ↑ 8.9%	17.3% ↓ 63.9%	16.0% ↓ 57.3%	15.0% ↓ 84.6%	99.3% ↓ 0.3%	31.1% ↓ 68.3%	43.3% ↓ 49.9%
SQA-IMG (PV : PT : DT = 65% : 30% : 5%)								
Baseline (No Specialization)	42.5%	74.1%	52.6%	49.0%	97.5%	99.7%	98.2%	94.9%
$\text{SMoES}_{\text{attention-soft}}$	8.4% ↓80.3%	76.3% ↑ 2.9%	30.1% ↓ 42.9%	41.7% ↓15.0%	33.2% ↓ 65.9%	99.3% ↓ 0.4%	54.3% ↓ 44.7%	90.0% ↓ 5.2%
$\text{SMoES}_{\text{gaussian-soft}}$	2.7% ↓ 93.6%	80.3% ↑ 8.2%	27.5% ↓ 47.7%	17.7% ↓ 63.9%	13.0% ↓ 86.6%	99.2% ↓ 0.5%	40.6% ↓ 58.7%	49.9% ↓ 47.4%
POPE (PV : PT : DT = 86% : 11% : 3%)								
Baseline (No Specialization)	42.9%	85.7%	47.8%	37.7%	97.8%	100%	98.1%	87.8%
$\text{SMoES}_{\text{attention-soft}}$	8.0% ↓ 81.4%	88.2% ↑ 2.9%	17.1% ↓ 64.1%	31.4% ↓ 16.7%	33.4% ↓ 65.9%	99.9% ↓ 0.1%	41.0% ↓ 58.2%	84.4% ↓ 3.8%
$\text{SMoES}_{\text{gaussian-soft}}$	1.8% ↓ 95.8%	91.5% ↑ 6.8%	12.0% ↓ 74.8%	14.5% ↓ 61.5%	9.0% ↓ 90.8%	99.8% ↓ 0.2%	19.4% ↓ 80.2%	40.8% ↓ 53.6%
GQA (PV : PT : DT = 86% : 11% : 3%)								
Baseline (No Specialization)	43.0%	85.5%	47.9%	38.3%	97.9%	100%	98.1%	89.2%
$\text{SMoES}_{\text{attention-soft}}$	8.2% ↓ 80.9%	87.7% ↑ 2.6%	17.4% ↓ 63.6%	32.1% ↓ 16.2%	34.0% ↓ 65.2%	99.9% ↓ 0.1%	41.7% ↓ 57.5%	84.8% ↓ 5.0%
$\text{SMoES}_{\text{gaussian-soft}}$	1.9% ↓ 95.7%	90.9% ↑ 6.4%	12.2% ↓ 74.5%	16.2% ↓ 57.7%	9.3% ↓ 90.5%	99.8% ↓ 0.2%	19.8% ↓ 79.8%	42.4% ↓ 52.5%
TextVQA (PV : PT : DT = 80% : 17% : 3%)								
baseline	43.6%	72.5%	48.6%	43.1%	97.9%	99.1%	98.1%	90.7%
$\text{SMoES}_{\text{attention-soft}}$	8.9% ↓ 79.5%	72.0% ↓ 0.8%	19.9% ↓ 59.0%	34.5% ↓ 19.9%	35.8% ↓ 63.5%	97.1% ↓ 2.1%	46.5% ↓ 52.6%	82.6% ↓ 8.9%
$\text{SMoES}_{\text{gaussian-soft}}$	2.6% ↓ 93.9%	77.2% ↑ 6.4%	15.6% ↓ 67.8%	21.0% ↓ 51.1%	12.6% ↓ 87.1%	97.5% ↓ 1.6%	27.4% ↓ 72.0%	47.1% ↓ 48.0%

for automotive applications. Given the high efficiency requirements for edge-side deployment, we test with a small-scale OLMoE-based VLM model with 7B parameters. Two NVIDIA Orin GPUs are connected via 10Gb Ethernet and deployed with Expert Parallelism (EP). Memory capacity is one of the bottlenecks for edge-side resources. Compared to Tensor Parallel (TP) and Data Parallel (DP) strategies, this approach has no redundancy in either weights or KV Cache. Compared to Attention-FFN separation with the same memory resource redundancy, this deployment has lower token transfer time, and memory space usage is relatively balanced. For the baseline deployment, since the experts are balanced, asynchronous transmission cannot give performance benefits and may even reduce efficiency, so synchronous transmission is adopted. For our SMoES with modality specialization, since there are more local experts with longer computation time, we adopted asynchronous transmission, allowing network transmission and expert computation to execute in parallel. Additionally, the optimized deployment is similar to PD separation, where

Table S11. TTFT (Time to First Token, Prefill) and TPOT (Time Per Output Token, Decode) speed improvement of $\text{SMoES}_{\text{attention-soft}}$ compared to soft-routing baseline. Δ: speedup percentage.

Benchmark	Method	Batch Size=1		Batch Size=2		Batch Size=4		Batch Size=8		Batch Size=16		Batch Size=32	
		TTFT(s)	TPOT(s)	TTFT(s)	TPOT(s)	TTFT(s)	TPOT(s)	TTFT(s)	TPOT(s)	TTFT(s)	TPOT(s)	TTFT(s)	TPOT(s)
MMMU	baseline	2.810	0.786	3.819	0.853	5.100	0.981	7.949	1.414	11.562	2.078	17.626	2.552
	SMoES	**2.519**	**0.703**	**3.303**	**0.767**	**4.302**	**0.890**	**6.203**	**1.287**	**8.804**	**1.899**	**13.638**	**2.309**
	Δ	↓10.3%	↓10.5%	↓13.5%	↓10.1%	↓15.7%	↓9.3%	↓22.0%	↓9.0%	↓23.9%	↓8.6%	↓22.6%	↓9.6%
SQA-IMG	baseline	1.493	0.766	2.940	0.858	4.217	0.951	5.824	1.278	9.759	2.100	15.036	2.721
	SMoES	**1.356**	**0.692**	**2.558**	**0.775**	**3.656**	**0.858**	**4.859**	**1.134**	**7.571**	**1.885**	**12.023**	**2.431**
	Δ	↓9.2%	↓9.7%	↓13.0%	↓9.7%	↓13.3%	↓9.8%	↓16.6%	↓11.3%	↓22.4%	↓10.3%	↓20.0%	↓10.6%

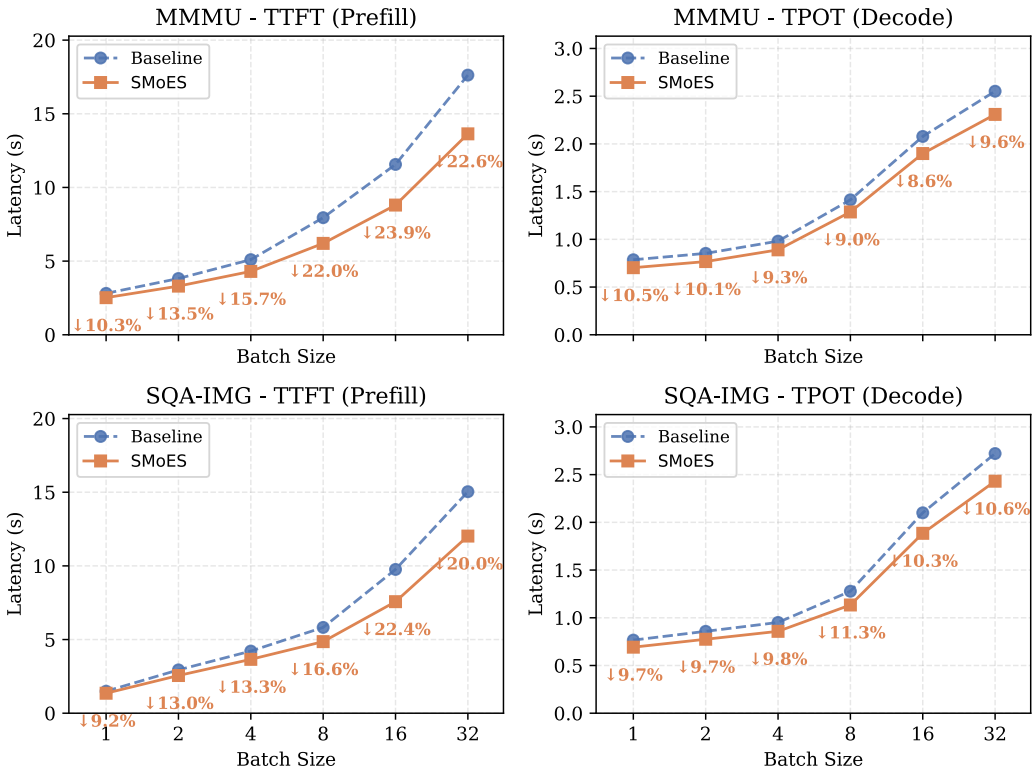

Figure S1. Latency decrease of SMoES compared to soft routing baseline at different batch sizes on edge-side deployment. The deployed VLM is based on OLMoE with 7B parameters. TTFT: Time to First Token (prefill stage); TPOT: Time Per Output Token (decode stage).

KV Cache transmission can be transferred asynchronously and be overlapped. The baseline uses a sequential order to assign experts, which is equivalent to a random order since there are no order constraints during training.

Based on the previous discussion, we can see that experts exhibit significant specialization. During actual deployment and data transmission, the data goes through DeDup (deduplication) to avoid transmitting duplicate tokens routed to experts on the same device in top-K routing. As shown in Tab. S10, we report the cross-GPU EP transfer ratio for vision and text tokens separately, since the number of vision and text tokens differs between prefill and decode phases. We also provide the overall proportion of transmitted tokens (V+T) at prefill stage. The table compares the transfer ratios with and without DeDup, demonstrating that our specialized methods (SMoES) significantly reduce the communication overhead for most of the cases, especially for vision tokens during the prefill stage. The conclusion is consistent among different benchmark datasets, indicating that our specialized methods (SMoES) are effective in reducing the communication overhead for most of the vision-language tasks.

The inference TTFT (Time to First Token) (Prefill) time and TPOT (Time Per Output Token) (Decode) time are reported in Tab. S11 and visualized in Fig. S1. The performance improvement mainly comes from the reduction in

communication time and the parallel execution of expert computation and communication, as evidenced by the latency breakdown in Fig. S1. For Prefill, as the batch size increases, the proportion of communication time increases, resulting in significant optimization performance, which aligns with the widening gaps observed in both Tab. S11 and Fig. S1. For Decode, when the Batch Size is small, due to the limited number of tokens, fewer experts are activated. Since the activated experts are proportional to the network transmission volume to a certain extent, the proportion of network transmission time remains constant, and thus the improvement ratio also remains constant.

2.4.1. Larger-Scale EP

Edge-side VLM/VLA deployment is crucial for autonomous driving and robotics. Scaling SMoES to cloud environments presents complex optimization challenges—where scheduling must account for multidimensional variables such as query modality ratios, expert specialization/redundancy, and scale-up/out bandwidth—yet it also unlocks greater efficiency potential. With proper optimization, SMoES's specialized expert groups allow converting costly cross-node communication into local computation, significantly reducing overhead compared to modality-agnostic baselines. While fully solving this complex scheduling is non-trivial, we evaluate a 16-GPU (2-node) scenario using a simple modality-based arrangement to reveal the substantial potential of SMoES. Without deep optimization, SMoES still reduces the inference time by 5.6% compared to the baseline (Tab. S12).

Batch size	1	2	4	8	16	32
Baseline	4.3	7.2	13.2	24.1	47.3	93.4
SMoES	4.1	6.7	12.1	23.1	45.4	88.2
Δ	-4.7%	-6.9%	-8.3%	-4.2%	-4.0%	-5.6%

Table S12. Inference time (s) of Qwen3-MoE (16 GPUs on 2 nodes).

2.4.2. Training Overhead

The training overhead is shown in Tab. S13. SMoES does not affect model parameters, except that the Gaussian Estimator adds a small buffer of approximately 60M. TFLOPS and peak GPU memory usage are almost unchanged. SMoES introduces a slight increase in training time, with the Gaussian Estimator contributing more due to global feature synchronization. But it becomes less noticeable as the model scales up. Note that SMoES introduces no additional overhead during inference.

2.5. Visualizations

We show routing distributions for SMoES based on the four backbones in Figs. S2 to S4. We can see that at most

Model	Params (B)	TFLOPS	Step Time (s)	Mem. (GB)
32 experts	1.19	1.82	0.74	22.3
+inter-bin MI-loss	1.19	1.82	0.80	22.3
+Attention Estimator	1.19	1.82	0.81	22.3
+Gaussian Estimator	1.25	1.83	0.93	22.3
64 experts	1.59	1.90	0.82	31.4
+inter-bin MI-loss	1.59	1.90	0.90	31.4
+Attention Estimator	1.59	1.90	0.91	31.4
+Gaussian Estimator	1.65	1.91	0.99	31.4

Table S13. Training costs on OLMoE for one GPU (batch 2, EP.8).

layers, there is one or several bins that are dedicated to vision tokens. In vision-language tasks, vision tokens typically dominate in quantity but have lower information density. By allocating dedicated experts to handle these abundant yet information-sparse vision tokens, other experts are freed up to process more complex text information. This is why SMoES achieves significantly better performance on text-only tasks compared to other methods.

However, vision tokens are not always low in information density; some vision tokens carry critical information. Fortunately, our modality specialization method is soft and dynamic, allowing the model to learn to route different vision tokens to different experts. This enables important vision tokens to be assigned to specialized experts for processing, preventing information loss. This is why our method outperforms manually pre-defined hard-routed and hybrid-routed approaches.

Except for Moonlight-MoE, which shows a relatively flat MSI layer-wise curve, most models exhibit the characteristic that shallow layers have higher modality specialization (MSI) while deeper layers show lower specialization, indicating a gradual fusion of modalities as the network goes deeper. For Moonlight-MoE, the additional router score bias makes its routing distribution less uniform, which slightly reduces its overall modality specialization level compared to other backbones.

We show the evolution of expert specialization during training on the four backbones in Figs. S5 to S9. At the early stage of training, the baseline and SMoES exhibit similar expert tendencies, with specialization scores roughly evenly spread across the spectrum. However, as training progresses, baseline experts tend to collapse toward the middle, meaning most experts become mixed-modal and handle both vision and text tokens simultaneously. In contrast, SMoES maintains pronounced modality specialization: one subset of experts separates to process vision-dominant tokens, another focuses on text-dominant tokens, and the remaining experts stay mixed to handle multimodal inputs. This specialized pattern is especially evident in shallow layers where vision and text tokens remain far apart in representation space; deeper layers, after repeated cross-modal

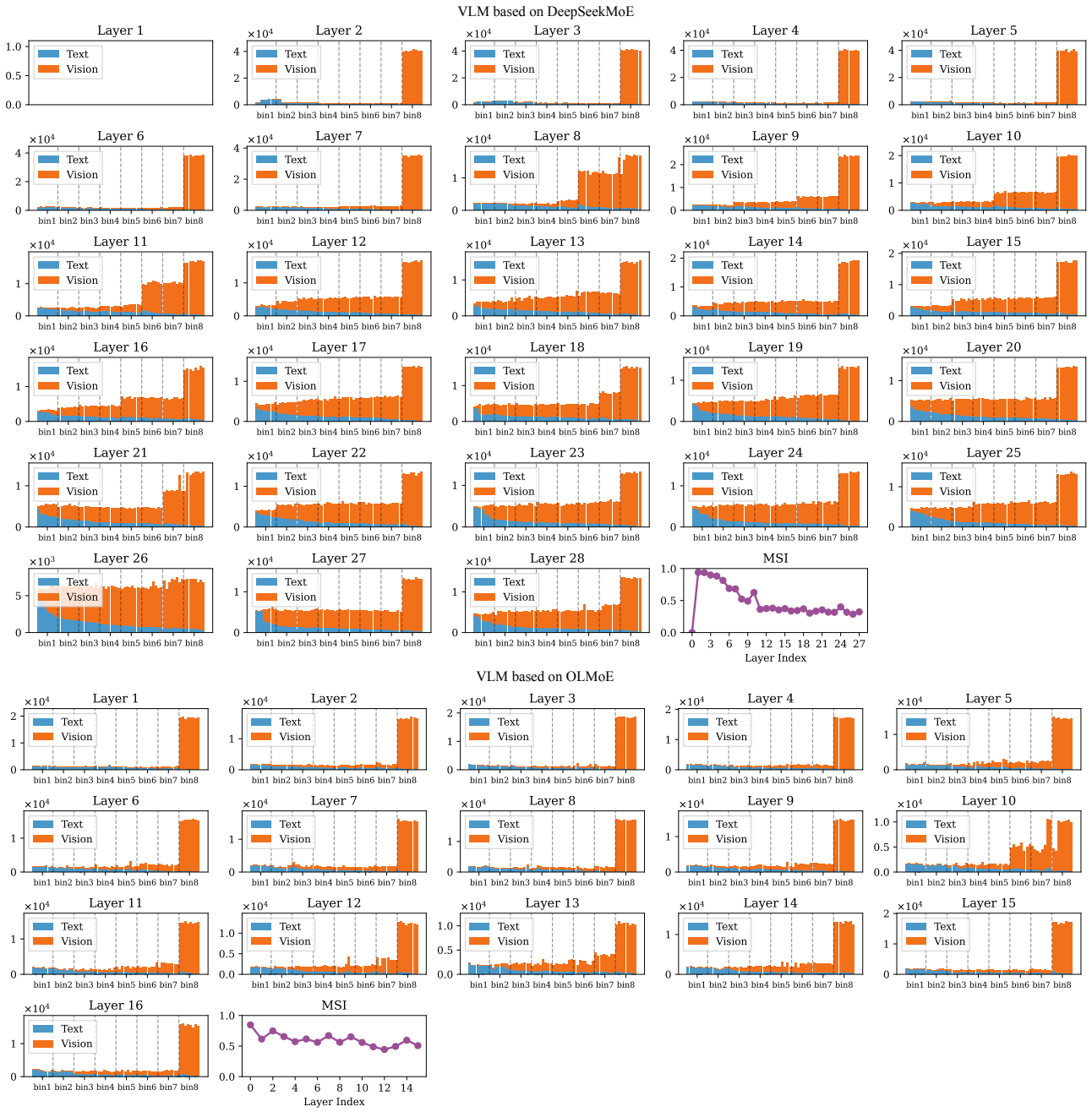

Figure S2. Routing distribution of tokens to experts in DeepSeekMoE and OLMoE. Horizontal axis: all experts grouped into eight expert bins. Vertical axis: number of tokens routed to each expert.

interactions, naturally become more multimodal, which explains the reduced but still observable specialization gap.

We show the modality fusion patterns for SMoES$_{\text{attention-soft}}$ in the four backbones in Figs. S10 to S12. Across models, shallow layers tend to produce soft modality scores near the two extremes (pure vision or pure text), and the scores gradually converge toward the center as depth increases. The convergence rate and shape vary with the backbone: DeepSeekMoE and Qwen3-MoE often split vision tokens into multiple peaks, Moonlight-MoE keeps a single peak. These differences indicate that backbone architecture and pretraining induce distinct token characteristics, while also showing that SMoES adapts to diverse MoE designs.

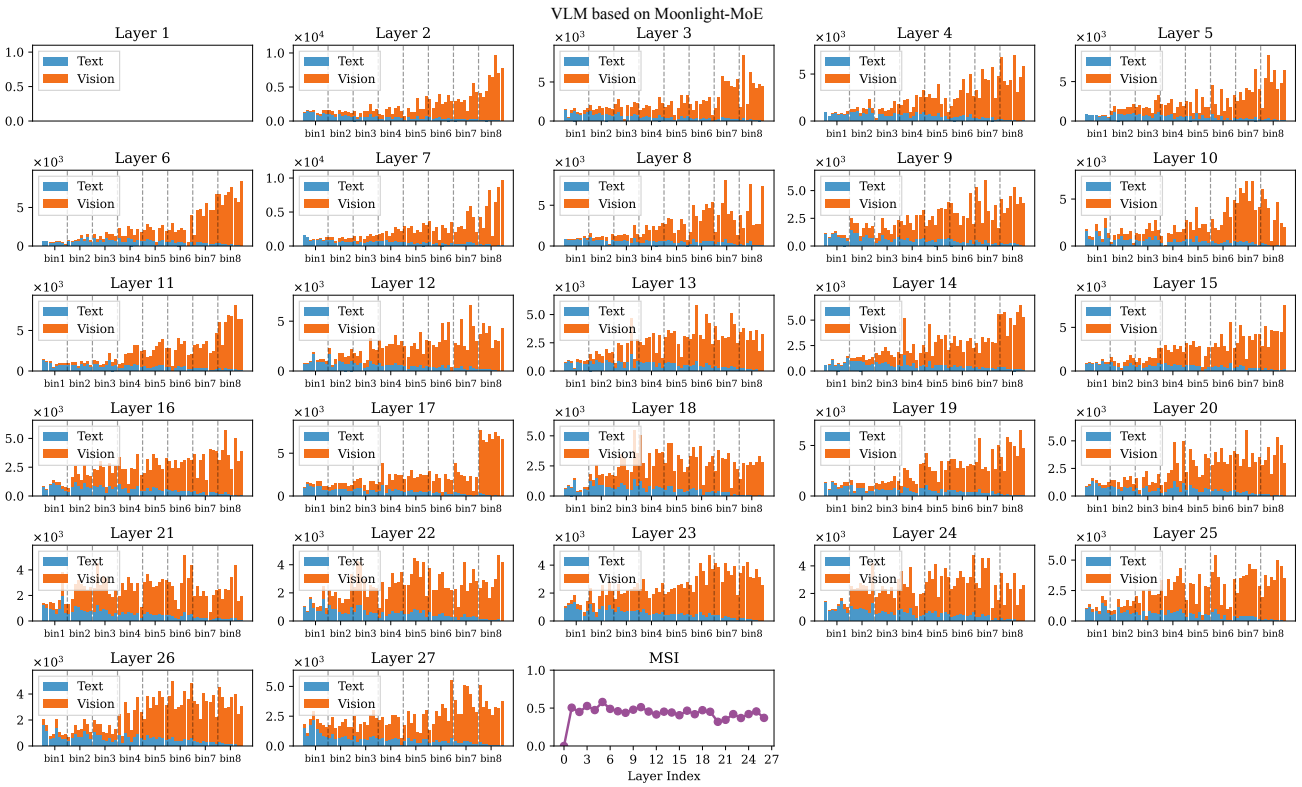

Figure S3. Routing distribution of tokens to experts in Moonlight-MoE. Horizontal axis: all experts grouped into eight expert bins. Vertical axis: number of tokens routed to each expert.

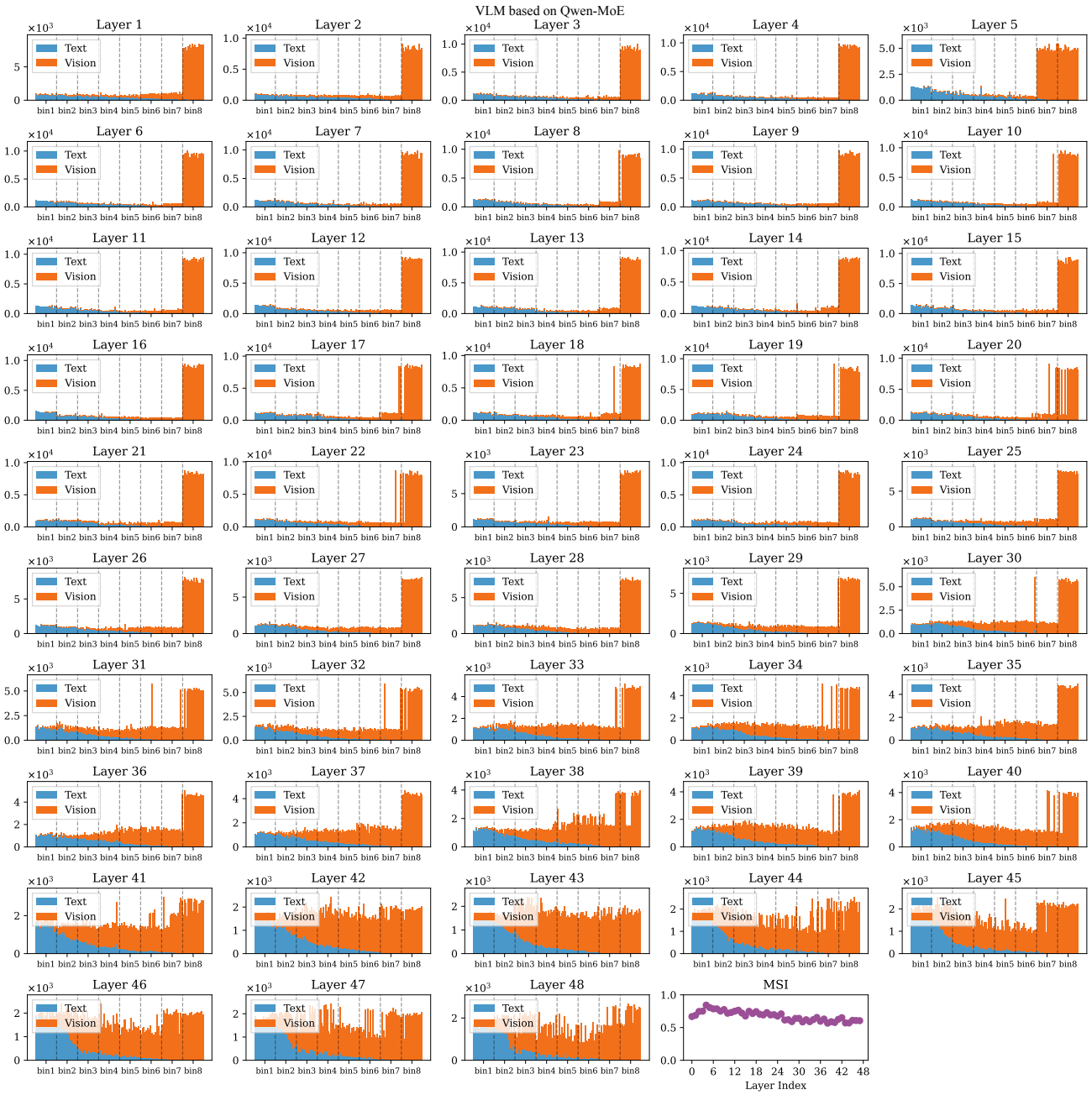

Figure S4. Routing distribution of tokens to experts in Qwen3-MoE. Horizontal axis: all experts grouped into eight expert bins. Vertical axis: number of tokens routed to each expert.

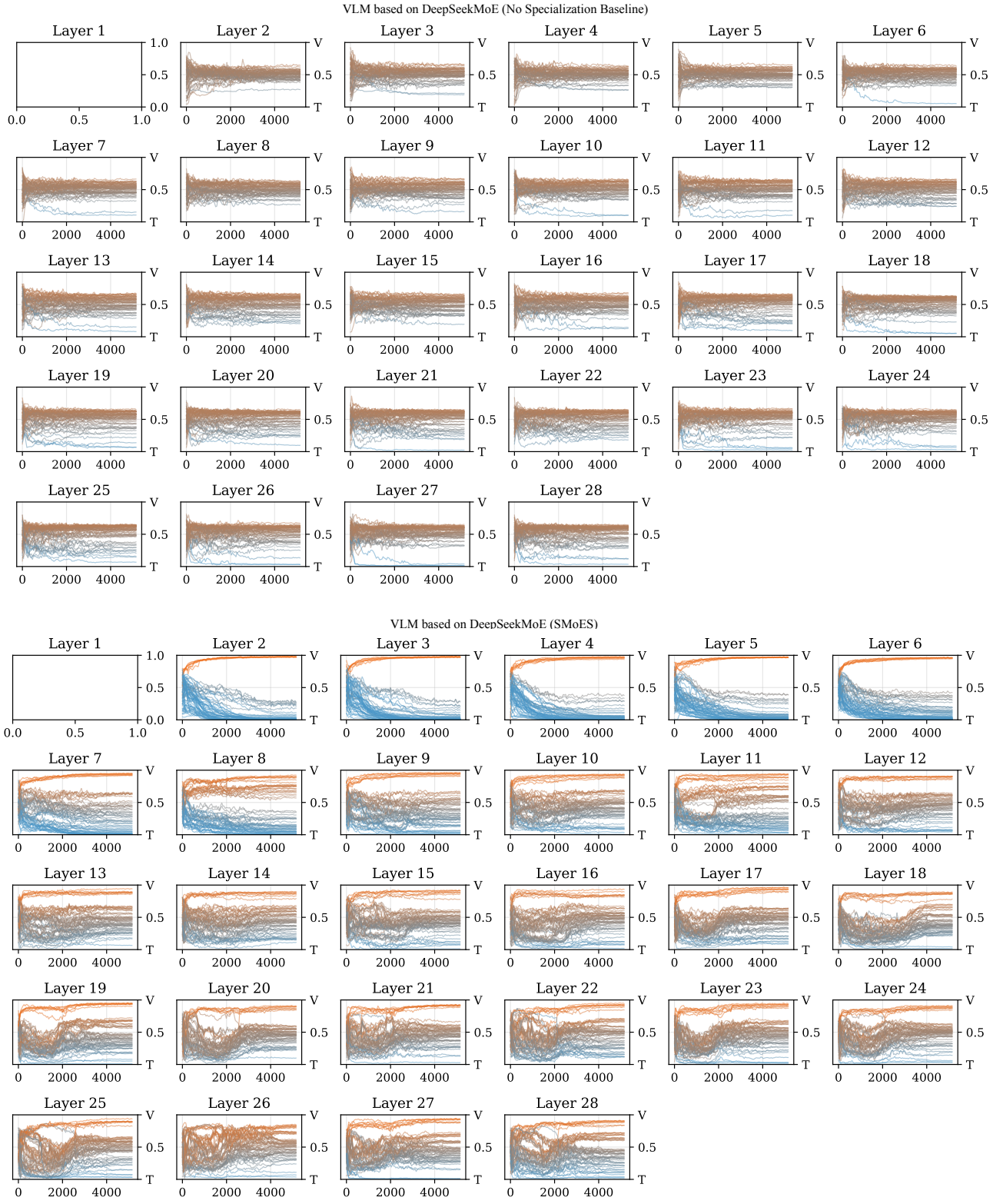

Figure S5. Evolution of expert specialization during training on DeepSeekMoE. Each curve represents an expert. Horizontal axis: training steps. Vertical axis: expert specialization score (symmetric expansion of MSI). V: vision specialization; T: text specialization.

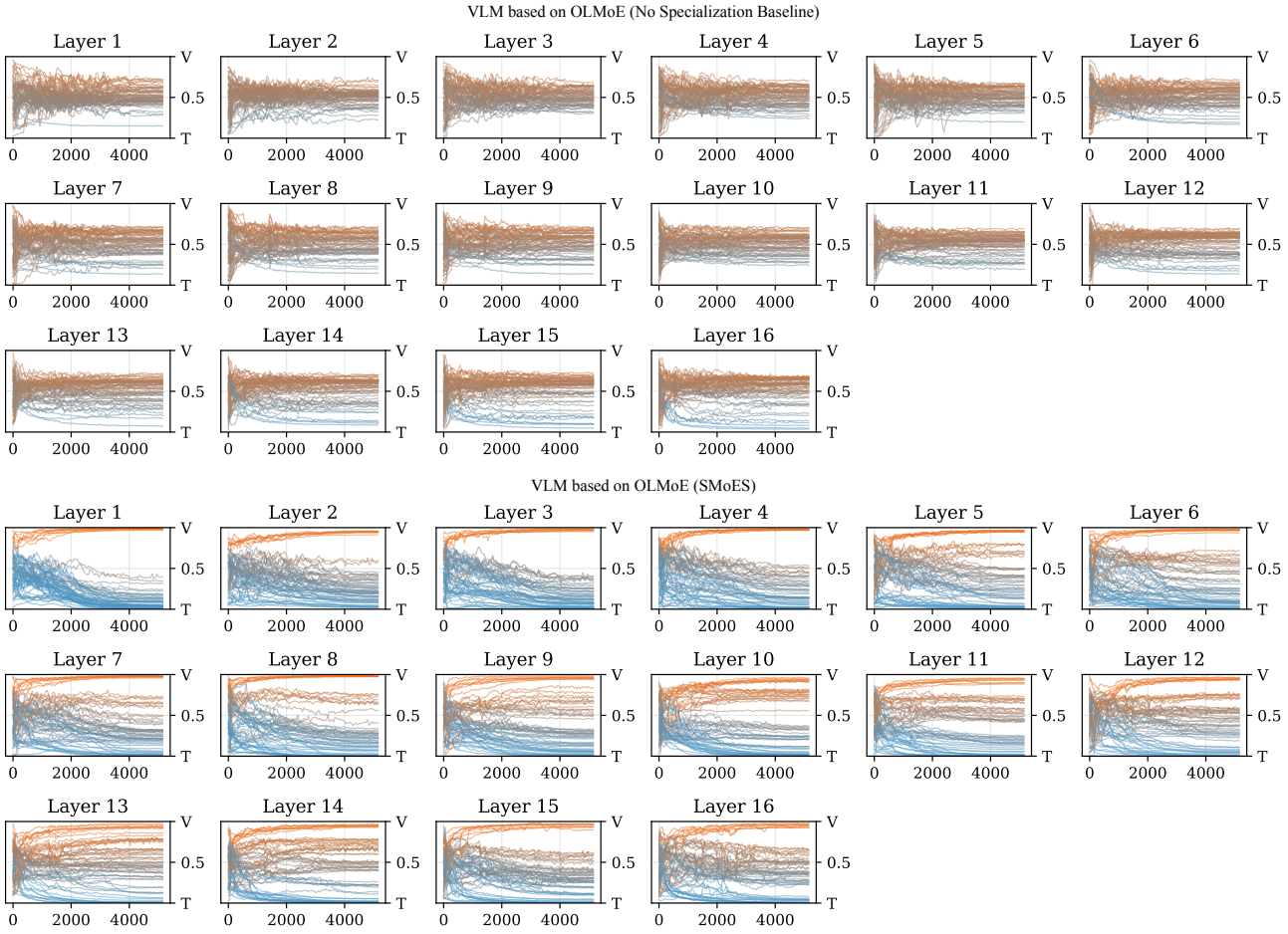

Figure S6. Evolution of expert specialization during training on OLMoE. Each curve represents an expert. Horizontal axis: training steps. Vertical axis: expert specialization score (symmetric expansion of MSI). V: vision specialization; T: text specialization.

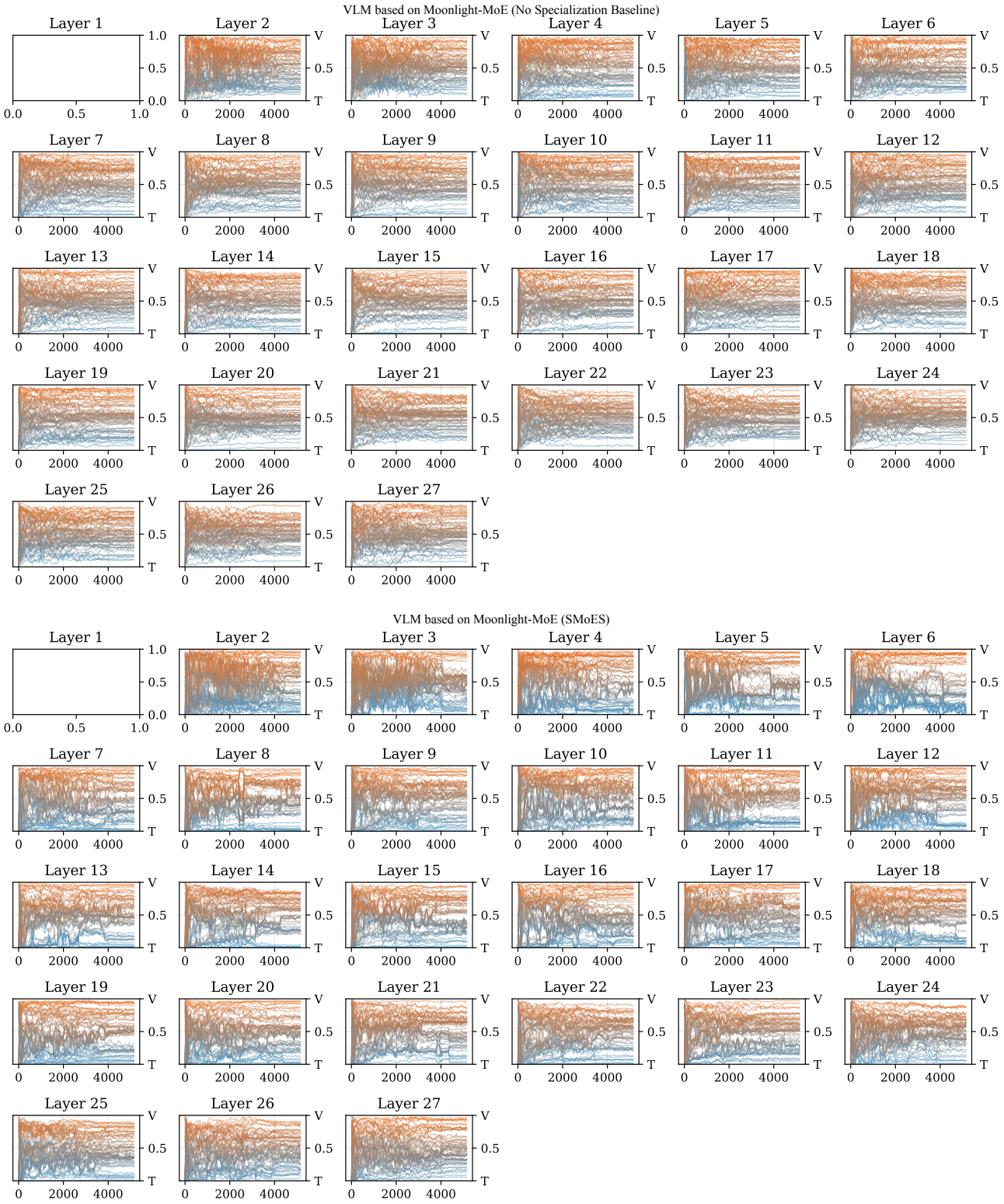

Figure S7. Evolution of expert specialization during training on Moonlight-MoE. Each curve represents an expert. Horizontal axis: training steps. Vertical axis: expert specialization score (symmetric expansion of MSI). V: vision specialization; T: text specialization.

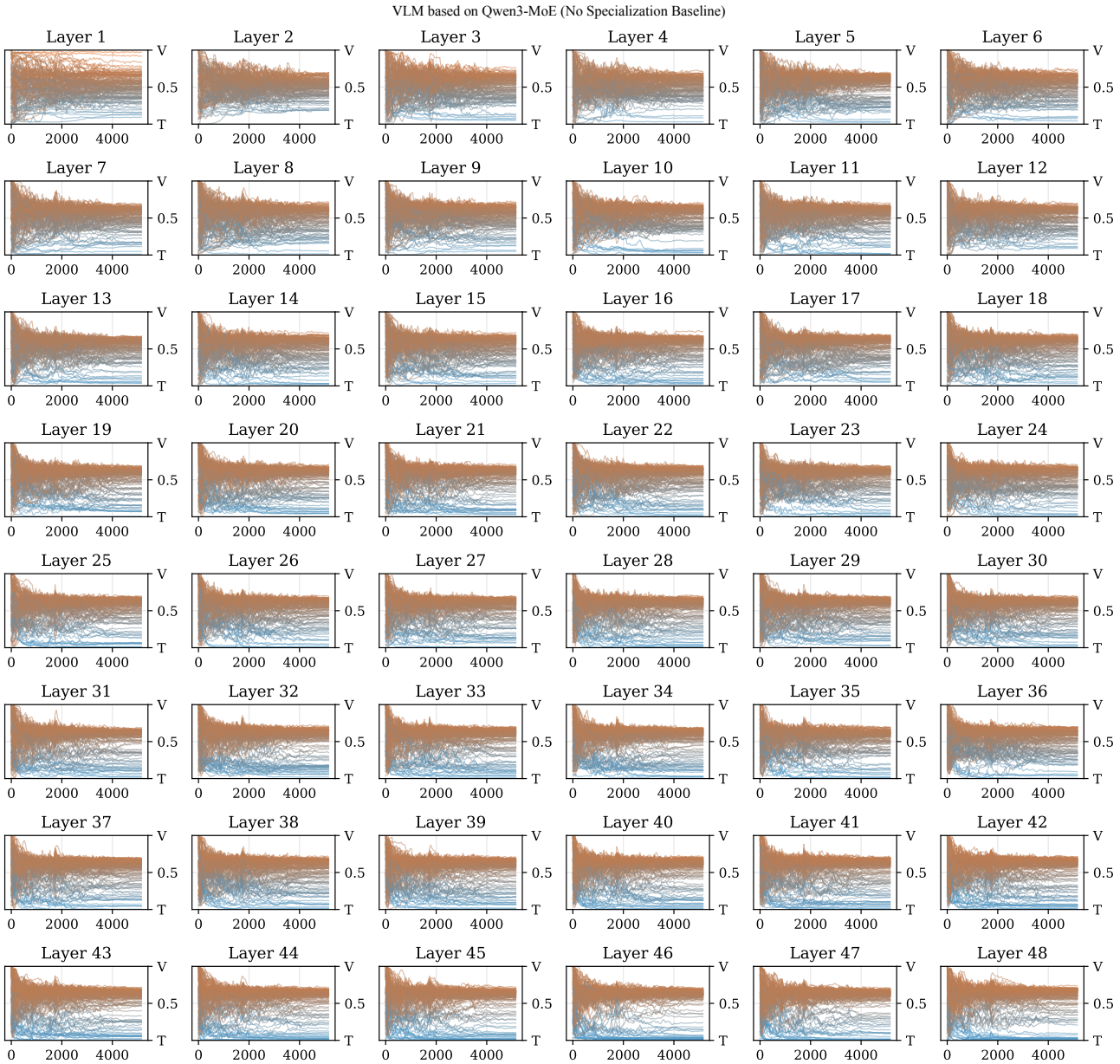

Figure S8. Evolution of expert specialization during training on Qwen3-MoE with no specialization (baseline). Each curve represents an expert. Horizontal axis: training steps. Vertical axis: expert specialization score (symmetric expansion of MSI). V: vision specialization; T: text specialization.

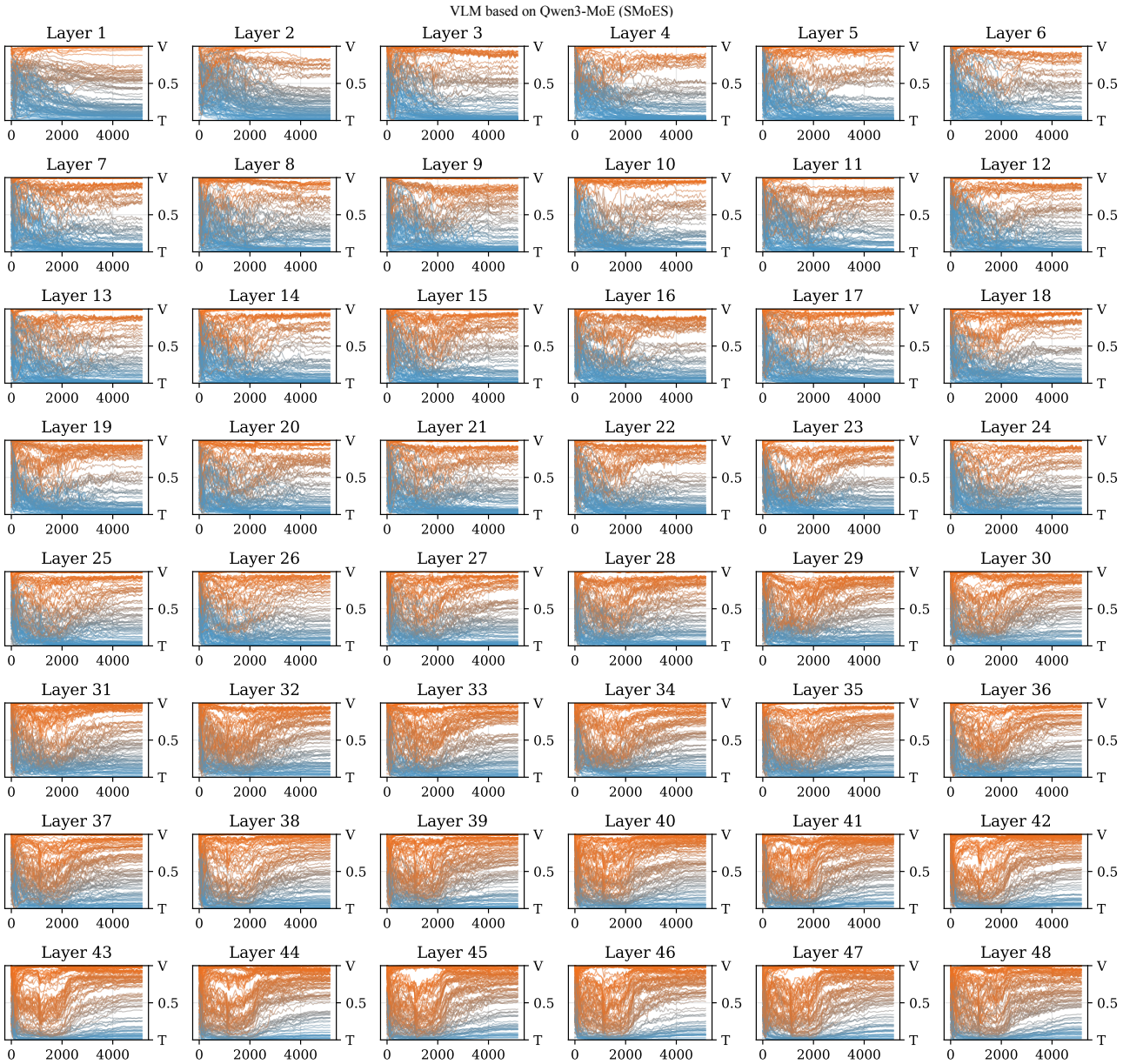

Figure S9. Evolution of expert specialization during training on Qwen3-MoE with SMoES. Each curve represents an expert. Horizontal axis: training steps. Vertical axis: expert specialization score (symmetric expansion of MSI). V: vision specialization; T: text specialization.

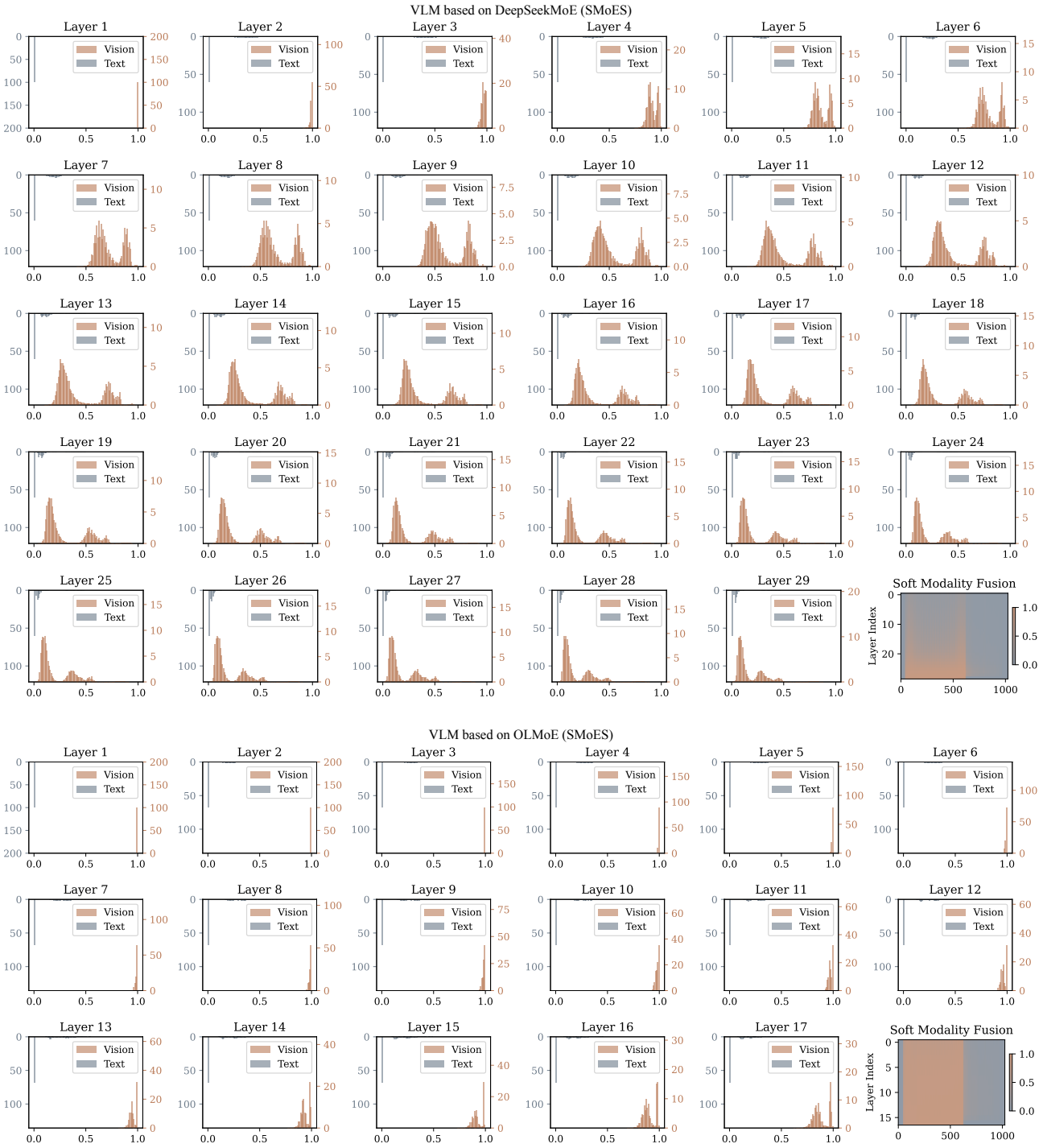

Figure S10. Modality fusion patterns for $\text{SMoES}_{\text{attention-soft}}$ in DeepSeekMoE and OLMoE. Layer subplots: horizontal axis is the soft modality score, and vertical axis is the number of token distribution. Soft modality fusion subplot: horizontal axis is the token id in a sample sequence, and vertical axis is the layer index.

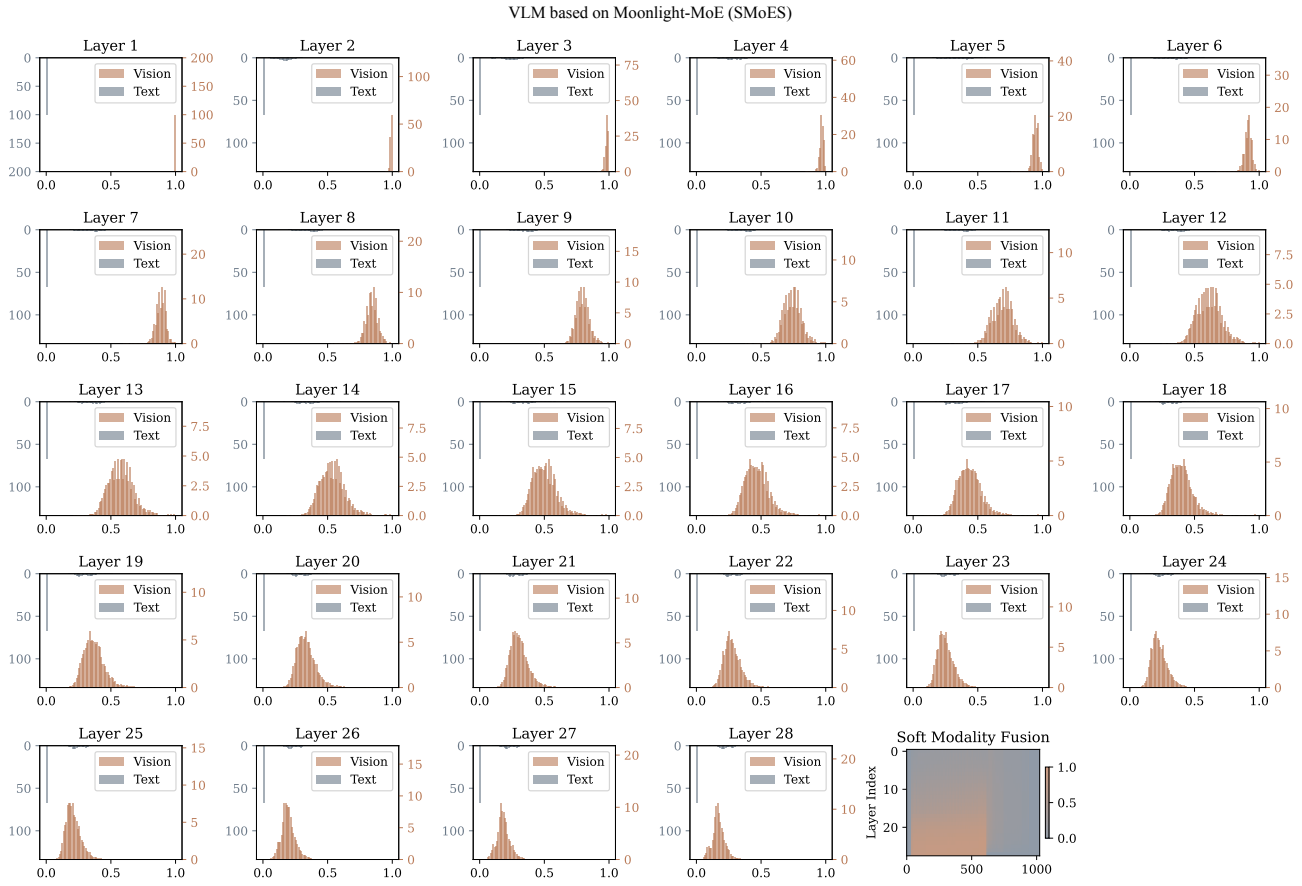

Figure S11. Modality fusion patterns for $\text{SMoES}_{\text{attention-soft}}$ in Moonlight-MoE. Layer subplots: horizontal axis is the soft modality score, and vertical axis is the number of token distribution. Soft modality fusion subplot: horizontal axis is the token id in a sample sequence, and vertical axis is the layer index.

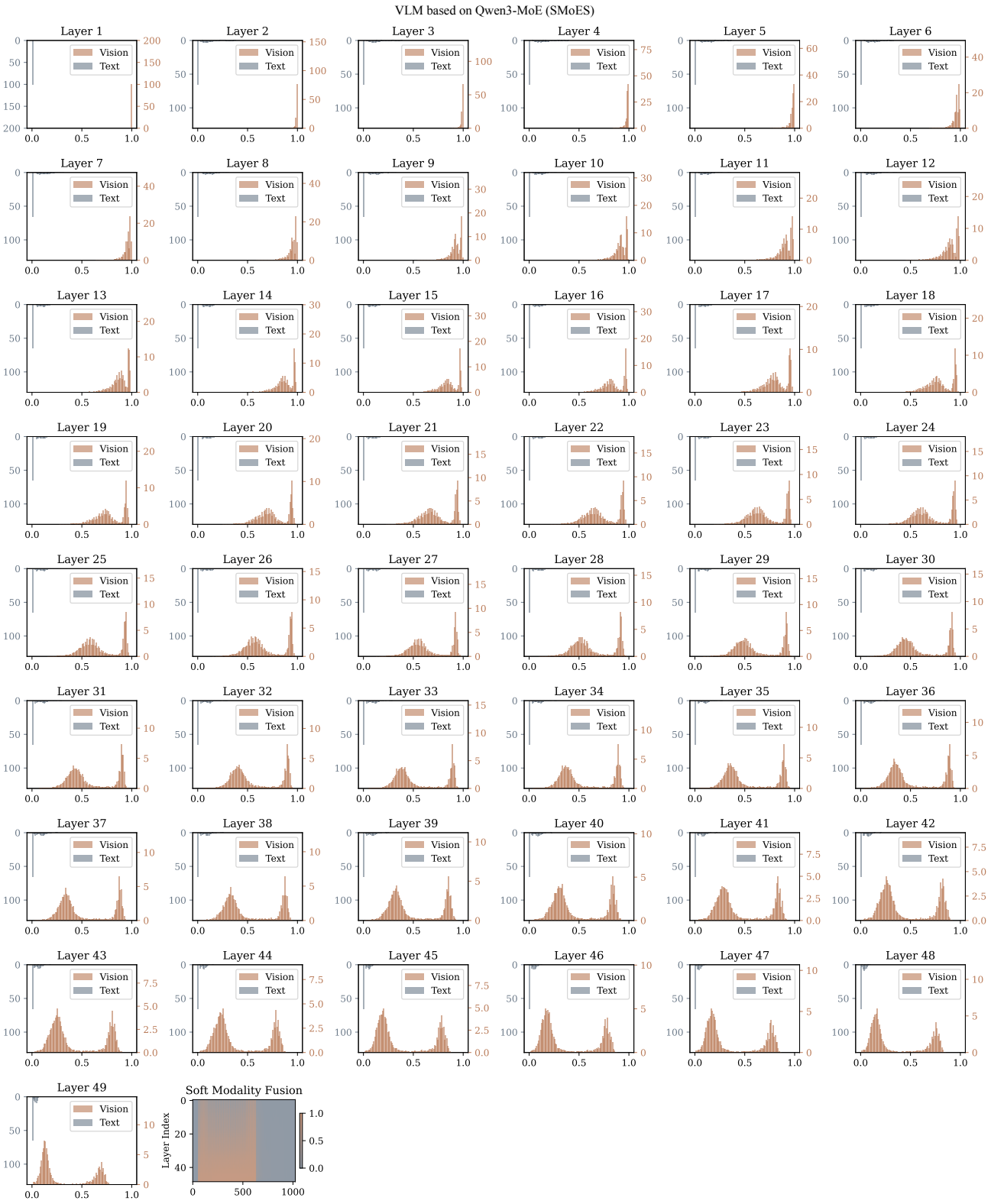

Figure S12. Modality fusion patterns for $\text{SMoES}_{\text{attention-soft}}$ in Qwen3-MoE. Layer subplots: horizontal axis is the soft modality score, and vertical axis is the number of token distribution. Soft modality fusion subplot: horizontal axis is the token id in a sample sequence, and vertical axis is the layer index.